Faster Writing Scales Science, Better Reading Crowds It: Topic Choice and the Economics of Duplication

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Abstract

Reading has two jobs in science. It *teaches*, raising the quality of new work, and it *protects*, revealing projects that others have already done. We model researchers who choose a topic and split their time between reading and writing. For any learning and detection technology with bounded absorption, the payoff of working in a topic is single peaked in the flow of its literature. Free choice of topics then has the structure of a system of cities: topics fill past their best size, and the duplication rate is pinned by entry rather than by the volume of papers. The implication we emphasise is an asymmetry. Faster writing raises the payoff of opening new topics as much as that of joining old ones, so it scales output almost one for one. Better reading, search and summarising tools raise only the payoff of existing topics, and entry competes the gain away. On arXiv, more volume opens new ground rather than thickening old topics. On 2.2 million submissions with exact timestamps, four arXiv rule changes and design features identify the margins the model relies on. A tighter endorsement rule cut first-time submitters by about a sixth. A peer-review requirement halved review articles in computer science. A cap of two submissions per person per month binds on about one paper in a hundred. Daily listing position barely moves readership and leaves the detection of overlapping work unchanged. In the calibrated model, a 30% faster pen raises output per researcher by 26.1 log points and 30% faster reading by 0.23. The static gain from optimal topic choice is small, and forward-looking entry still makes fast-growing topics overshoot.

JEL: O31, O33, O40, D62, I23, R12. **Keywords:** economics of science, topic choice, duplication, system of cities, burden of knowledge, artificial intelligence, arXiv.

1 Introduction

Submissions to arXiv more than doubled between 2018 and 2025, and in September 2026 the archive received a record 40,363 papers. Large language models now draft text faster than any researcher can, and they are starting to read and summarise for us too. Two opposite hopes and fears follow. Faster writing might flood science with duplicates. Better reading tools might be the remedy and raise research productivity. This paper argues that free choice of research topics turns both intuitions around. Faster writing scales science, while better reading tools are largely competed away.

The argument rests on one observation: researchers choose not only how much to read and write but *where* to work. Reading has two jobs. It *teaches*: what one reads raises the quality of one's own work, and more so in a topic with more to learn from. It *protects*: each earlier paper may already cover one's project, and reading is how one finds out. Unaware duplicates lose part of their credit; detected overlaps can be turned into follow-ons that cite the earlier work, or abandoned. These two jobs make the payoff of working in a topic rise with the flow of its literature while there is much to learn, and fall once more of that flow goes unread. Newcomers enter topics until they are no better than the frontier, where new topics are opened.

The headline result. A tool that speeds up writing raises the payoff of everything a researcher does, including opening a new topic. Newcomers spread out, so output per researcher rises almost one for one with writing speed, and the duplication rate barely moves (Propositions 7 and 8). A tool that speeds up reading, search or catching up on a literature raises the payoff only of working inside existing topics. Newcomers crowd in until those topics are again no better than the frontier. Density rises, fewer topics are opened, and output per researcher rises only to the extent that the frontier improves as fewer directions are taken. The gain from better reading is dissipated by entry in the same way that a better road into a city is capitalised into its size. This asymmetry gives one reason why decades of better access to the literature have coincided with falling measured research productivity (Bloom et al., 2020).

What is new, and what is not. The equilibrium structure is that of a system of cities (Henderson, 1974) or of clubs (Buchanan, 1965). The average payoff is single peaked in group size, new groups enter freely, equilibria are stable only on the falling side, the planner sits at the peak of the average payoff, and an entry tax decentralises the optimum. Self-organised cities are too large for the same reason (Henderson and Becker, 2000; Scotchmer, 2002). We use this structure and claim four things that it does not provide. First, a microfoundation of the hump in the two jobs of reading, valid for any learning and detection technology in a broad class. The proof uses joint log-concavity of the researcher's problem in reading time and density (Theorem 1), and the hump comes from bounded absorption against a covering hazard that grows with the literature. Second, the crowders are the readers: the sign of the spillover from one more paper is derived from the envelope theorem rather than assumed, and it changes along the payoff curve (Proposition 1).

Third, the duplication rate is pinned by entry, with an exact volume elasticity of density built from observables (Proposition 3). Fourth, the writing–reading asymmetry, together with a sufficient statistic for the stepping-on-toes parameter of growth theory (Proposition 10). A catch-up cost, the need to read a topic's stock before contributing (Jones, 2009), adds a wedge that the city literature lacks: incumbents' returns rise with density even where newcomers find topics too crowded (Corollary 1).

Evidence. We confront the model with the arXiv record in two ways. The descriptive evidence (Section 7) comes from 1.49 million papers posted in 2018–2025, classified into 150 text-based areas. The anchor fact is that volume opens new ground. From the 95th to the 25th percentile similarity radius, local density rises with area volume at an elasticity of 0.12–0.19, against 0.53–0.81 if papers kept landing in the same idea space. At the closest-overlap radius the estimate is 0.035, though it is imprecise (SE 0.118). Returns are hump-shaped in topic density. Unaware duplicates lose about a fifth of their citations, close to the 21% penalty for scooped teams in Hill and Stein (2025a), while aware follow-ons gain. Once corrected for citation recall, about 60% of close overlap is aware.

The quasi-experimental evidence (Section 8) uses arXiv's own rules on a new panel of 2.2 million arXiv submissions with exact timestamps through 15 October 2026. Four shocks speak to the margins the model relies on.

- *Entry cost.* In January 2026, arXiv stopped endorsing new submitters on the strength of an institutional e-mail address alone; Mathematics had moved in December 2025. The share of papers by first-time submitters fell by 3.4 percentage points (about 15%) in the affected sections, net of the year before. The staggered timing matches the rule.
- *Catch-up cost.* From November 2025, review and position papers in computer science needed prior peer review. Review-type papers in computer science halved relative to other archives, more so in categories that had relied on them. First-time authorship fell more in those categories, though a smaller pre-trend of the same sign makes this suggestive.
- *Writing cap.* From October 2026, each person may submit at most two papers a month. The cap removed about 1.3% of papers. Total volume kept growing at its usual seasonal pace, so per-submitter caps are a weak lever on volume.
- *Readership.* A paper's position in the daily listing depends on how many others submitted earlier that day. We use this to test the detection channel directly. Position has only a small effect on citations, and no detectable effect on later overlapping work or on the share of it that cites the paper, with tight bounds. Overlap is detected by targeted search rather than by browsing.

At the author level, adopters of LLM-style writing raised their output by about 7% relative to non-adopters. That is far less than the 36% of Kusumegi et al. (2025), whose design has been criticised for mechanically generating positive dynamics (Renault et al., 2026).

Magnitudes. The calibrated model is disciplined by the corrected arXiv moments (Section 9). The reading–writing asymmetry is the quantitatively robust prediction: a 30% faster pen raises output per researcher by 26.1 log points, while 30% faster reading raises it by only 0.23. The static welfare cost of crowding is small. Optimal topic choice would raise output per researcher by 0.5% in the baseline calibration and 3.3% in the within-topic one, and overlap destroys only about 1.7% of credit at mature density. The frictionless benchmark, in which catch-up is free and topics fill until the whole rent at the peak is dissipated, would lose 12%; it is a counterfactual, not the calibrated economy. The direction of the inefficiency survives treating all priority credit lost to overlap as a transfer between racers (Proposition 6). With the measured volume elasticity, topic crowding explains little of the decline in research productivity.

Implications. Two further results follow, and we collect them in Section 10. Because the stock of literature lags its flow, topics whose literature grows faster than a threshold rate overshoot their best density. This survives forward-looking entry: rational entrants overshoot more than myopic ones once a topic grows fast, because what deters entry is the stock, and a young topic has little of it (Proposition 9). And the stepping-on-toes parameter equals one minus the product of the volume elasticity of density and the slope of the entrant's payoff curve.

Related literature. The paper joins five literatures. The first is the economics of science and priority (Merton, 1957, 1961; Dasgupta and David, 1994; Stephan, 1996; Hill and Stein, 2025a,b; Bobtcheff et al., 2017) and the philosophy of the division of cognitive labour (Kitcher, 1990; Strevens, 2003). We add reading as a choice with two distinct roles, and derive rather than assume the costs of duplication. Hill and Stein (2025b) show that priority races lower quality; our cost of crowding works through detection and catch-up instead, and the two are complementary. The second is the theory of excess entry, common pools, cities and clubs (Gordon, 1954; Weitzman, 1974; Mankiw and Whinston, 1986; Henderson, 1974; Buchanan, 1965; Henderson and Becker, 2000; Scotchmer, 2002) and of patent races (Loury, 1979; Dasgupta and Stiglitz, 1980). Our resource is attention to a literature, and the crowders are its readers. The third is the direction of research (Bryan and Lemus, 2017; Hopenhayn and Squintani, 2021; Carnehl and Schneider, 2025; Bramoullé and Saint-Paul, 2010; Rzhetsky et al., 2015; Foster et al., 2015; Myers, 2020; Borjas and Doran, 2012, 2015; Azoulay et al., 2019). Carnehl and Schneider (2025) model the choice of question by novelty and feasibility. We model the choice by density, derive the hump, and measure the margin. Borjas and Doran (2015) document that researchers move in idea space in response to crowding, which is the margin our model turns on. The fourth is the macroeconomics of ideas and of access to knowledge (Romer, 1990; Aghion and Howitt, 1992; Jones, 1995; Kortum, 1997; Weitzman, 1998; Bloom et al., 2020; Jones, 2009; Ekerdt and Wu, 2024; Furman and Stern, 2011; Murray et al., 2016; Iaria et al., 2018; Biasi and Moser, 2021; Evans, 2008; McCabe and Snyder, 2015; Agrawal et al., 2016; Park et al., 2023). We give stepping on toes an equilibrium foundation and explain why better access to the literature need not raise productivity. The fifth is artificial intelligence in science (Agrawal et al., 2019; Brynjolfsson et al., 2019; Jones, 2025; Duede et al., 2026; Hao

et al., 2026; Liang et al., 2024; Kusumegi et al., 2025). Jones (2025) studies AI in a task-based R&D production function, and Duede et al. (2026) model LLMs as labour-augmenting technology that shifts effort across projects. We separate writing, entry, reading, search and catch-up channels by their equilibrium footprints. Methodologically, our text-based density measures follow Hoberg and Phillips (2016) and Kelly et al. (2021), and the listing-position design builds on Haque and Ginsparg (2009, 2010) and Feenberg et al. (2017).

Section 2 sets up the model and Section 3 derives the payoff curve. Sections 4 and 5 characterise equilibrium, welfare and social value. Section 6 states the writing–reading asymmetry. Sections 7 and 8 present the descriptive and quasi-experimental evidence, Section 9 quantifies the model, Section 10 collects the implications for fast-growing topics and growth, Section 11 lists out-of-sample predictions, and Section 12 concludes.

2 Model

The economy is stationary. Time is continuous, but within a topic the relevant literature is a flow, so we describe a researcher's choices per unit of time.

2.1 Researchers, topics and time

There is a continuum of topics. Topics that are open are ex ante identical. A mass N of researchers each has one unit of time per period, split into a share x spent writing and a share $r = 1 - x$ spent reading. Writing full time yields a papers (projects) per period. Topic j hosts n_j researchers who write, and receives a flow of

$$P_j = a\, x\, n_j \tag{1}$$

papers per period. P_j is the topic's *density*. We call the ratio of writing speed to reading speed, a/b, the *pace* of science.

2.2 Reading teaches

A researcher who reads full time reads b papers per period, so a reading share r yields $R = b\, r$ papers drawn from her topic's flow. What she reads raises the quality of her own work by the factor $Q(R, P)$.

Assumption 1 (Learning)**.** $Q : \mathbb{R}^2_+ \to [1, \infty)$ is continuous, twice continuously differentiable on $\mathbb{R}^2_{++}$, nondecreasing in both arguments, with $Q(0, P) = Q(R, 0) = 1$ and $\bar{Q}(R) \equiv \sup_P Q(R, P) < \infty$ for every R. Moreover $\ln Q$ is jointly concave and supermodular on $\mathbb{R}^2_+$.

Nothing is learned without reading or without a literature to read, and what one can learn is bounded by what one reads. Concavity of $\ln Q$ rules out increasing returns to reading. Supermodularity says that reading is worth more in a topic with more to learn from. Our running example is

$$Q(R, P) = \big(1 + \beta\, h(R, P)\big)^{\gamma}, \qquad h(R, P) = \frac{RP}{R + P}, \qquad \beta > 0,\ \gamma \in (0, 1]. \tag{2}$$

The harmonic aggregator h is close to R when the reader samples a small part of a rich literature and close to P when she reads a thin literature exhaustively. Because h is concave and homogeneous of degree one, the example satisfies Assumption 1 (Appendix A).

2.3 Reading protects

Each paper in the flow *covers* a given project independently with a small probability. The number of covering papers is therefore Poisson with mean θP. A covering paper is caught only if it is read. With reading coverage $c = R/P$ it is caught with probability $\varphi(c)$.

Assumption 2 (Detection)**.** $\varphi : \mathbb{R}_+ \to [0, 1]$ is continuous and twice differentiable on $(0, \infty)$, with $\varphi(0) = 0$, $\varphi' > 0$ and $\varphi'' < 0$.

Concavity says that the first papers one reads are the most informative about overlap. Random reading with replacement gives $\varphi(c) = 1 - e^{-c}$. Our example $\varphi(c) = 1 - e^{-\sigma c}$ indexes *search precision* by σ: a better search engine finds covering papers sooner.

Credit. Overlap erodes credit multiplicatively. Each covering paper that goes unread turns part of the project into a duplicate and removes a share λ of its credit. When a covering paper is caught, the researcher chooses between two options. She can *pivot* to a fresh variant at a time cost that leaves a share $1 - \kappa_p$. Or she can *proceed* as a follow-on that cites the earlier paper and keeps a share δ drawn from a distribution G. She proceeds if $\delta > 1 - \kappa_p$. The expected share kept after a detection is $1 - \ell \equiv \mathbb{E} \max\{\delta, 1 - \kappa_p\}$.

Assumption 3 (Credit)**.** $0 \leq \ell < \lambda \leq 1$ and $\theta > 0$.

Detection is valuable ($\ell < \lambda$), and duplicates need not earn zero ($\lambda < 1$ is allowed). The arXiv evidence (Fact 3) requires both. Credit here is private. Section 5 separates it from social value, since part of the credit lost to overlap may be a transfer to the earlier paper. Let $\Delta \equiv \lambda - \ell$ and $q \equiv 1 - G(1 - \kappa_p)$, the share of detections that proceed.

Lemma 1 (Credit and overlap)**.** *Under Assumptions 2 and 3, expected credit per project is*

$$C(R, P) = \exp\Big\{ -\theta P\big[\lambda - \Delta\, \varphi(R/P)\big] \Big\}. \tag{3}$$

*Covering papers that the researcher does not pivot away from (*flagged overlap*) arrive at rate* $\theta P\,[1 - (1 - q)\varphi]$*. A share*

$$A = \frac{q\,\varphi}{1 - (1 - q)\,\varphi} \tag{4}$$

of them is aware*: the later paper cites the earlier one. Unaware duplicates keep a share* $1 - \lambda$ *of credit. Aware follow-ons keep* $\mathbb{E}[\delta \mid \delta > 1 - \kappa_p]$*, which can exceed one.*

The lemma follows from Poisson thinning. Detected and undetected covering papers are independent Poisson counts with means $\theta P \varphi$ and $\theta P(1 - \varphi)$, and $\mathbb{E}[t^K] = e^{-\mu(1-t)}$ for $K \sim \text{Poisson}(\mu)$.

Three features matter below. Only $\theta\lambda$ and Δ/λ enter payoffs. Aware follow-ons are selected on high δ, which is why they can out-earn comparable papers. And the measured overlap flag is a fixed share of covering events. Its level is a normalisation; its changes are not.

2.4 The researcher's problem

A researcher in a topic of density P earns credit $a\,x\,Q(R,P)\,C(R,P)$ per period. Writing x projects is worth x, and each is worth QC. Dividing by the common factor a,

$$g(r,P) \equiv (1-r)\,Q(br,P)\,C(br,P), \qquad v(P) \equiv \max_{r\in[0,1)} g(r,P). \tag{5}$$

The *payoff curve* v is the central object of the paper. Writing speed a scales every payoff and so drops out of v. The researcher takes P as given: she ignores both that her papers teach her colleagues and that they add to what her colleagues must read.

Remark 1 (Topic size and what density measures). A topic has a fixed set of projects: the coverage rate θ is a property of the topic and does not shrink when the topic attracts more people. This is a normalisation, not a restriction, once density is measured locally. Let a field of breadth B consist of B cells of unit breadth, each receiving a flow P/B, and let researchers read and be covered within their own cell, so that the coverage rate per paper is θ_0/B. The field is then B independent copies of a unit topic with density P/B, and every result below holds with P replaced by P/B. The model's topic is therefore a neighbourhood of fixed size in idea space, and P is papers per unit of idea space: the quantity that nearest-neighbour similarity measures. If papers fall at rate P/B in a d-dimensional space, the distance to the k-th nearest earlier paper scales as $(kB/P)^{1/d}$. "Opening a topic" means moving into an empty cell, and the question of whether topics broaden as they grow becomes the question of how the volume elasticity varies with the similarity radius, which Section 7 measures.

2.5 The frontier, careers and catch-up

Frontier. Instead of joining an open topic, a researcher may explore new ground. An explorer opens a viable topic with probability $\pi(M)$, where M is the measure of topics already open. If she succeeds, her output is worth f_0 per unit of writing speed: the founding papers of a field. The frontier pays $a\,F(M)$ in expectation, with $F(M) = f_0\,\pi(M)$. We assume π is continuous, positive and nonincreasing, with elasticity $\eta \equiv -M\pi'(M)/\pi(M) \geq 0$: as more directions are taken, a fruitful new one is harder to find (Kortum, 1997; Bloom et al., 2020). Opening M topics takes $\Phi(M) = \int_0^M \mathrm{d}m/\pi(m)$ researchers. We assume $F(M) < \hat{v}$, the peak of the payoff curve defined below, so open topics are worth joining. A newly opened topic attracts followers only if a thin topic beats the frontier, $F < v(P_0)$, where P_0 is the flow of a newborn topic.

Careers and catch-up. Researchers leave their topic at rate ρ (retirement, a change of field) and are replaced by newcomers who choose a topic. The literature a newcomer must master is a

Table 1: Notation

a, b	papers written / read per period at full time	P	topic density (flow of papers per period)
x, r	writing and reading shares, $r = 1 - x$	$R = br$, $c = R/P$	papers read; reading coverage
$Q(R, P)$	quality gain from what is read	$\varphi(c)$	probability a covering paper is caught
θ	coverage rate per earlier paper	λ, ℓ	credit lost per unread / caught covering paper
$g(r, P)$, $v(P)$	payoff per unit of a; its maximum	$\hat{P}$, $\hat{v}$	peak of v and its height
$F(M)$, $\pi(M)$	frontier payoff; explorer's success probability	η	elasticity of π in topics opened M
ρ, δ	career turnover; decay of the stock's relevance	S, b_c	stock of literature; catch-up reading speed
$\kappa = \rho/(\delta b_c)$	catch-up cost per unit of density	$w(P)$	newcomer's career payoff, $v(P)e^{-\kappa P}$
P^*, P^m	equilibrium (mature) density	$\mathcal{E}$	elasticity of density with respect to N
g, g^*	topic diffusion rate; overshooting threshold	λ_J	stepping-on-toes parameter of Jones (1995)
λ_s, τ	social loss per unread covering paper; transfer share	$U(P)$	entrant's expected payoff along the consolidation path

stock S that accumulates the flow and loses relevance at rate δ: $\dot{S} = P - \delta S$, so $S = P/\delta$ in a stationary topic. A newcomer must read it at speed b_c before contributing. This is the *catch-up* cost: a topic-level version of the burden of knowledge of Jones (2009). Explorers face no stock.

Lemma 2 (Career value)**.** *With exponential careers, a newcomer who joins a stationary topic of density P and then earns $a\,v(P)$ per period obtains, relative to an explorer, the expected career value $w(P)/F(M)$, where*

$$w(P) \equiv v(P)\, e^{-\kappa P}, \qquad \kappa \equiv \frac{\rho}{\delta\, b_c}. \tag{6}$$

In a stationary topic a share $1 - e^{-\kappa P}$ of members are catching up, so $w(P)$ is also output per member and the topic has $n(P) = P\, e^{\kappa P}/(a\, x^*(P))$ members, where $x^*(P)$ is the optimal writing share.

Catch-up takes $T = S/b_c = P/(\delta b_c)$ periods, and a career survives it with probability $e^{-\rho T} = e^{-\kappa P}$. The frictionless benchmark is $\kappa = 0$ ($b_c \to \infty$). Incumbents earn $a\,v(P)$ per period; newcomers earn $a\,w(P)$ on average over their careers.

2.6 Equilibrium

Definition 1 (Topic equilibrium)**.** A stationary topic equilibrium is a density P^* for every open topic, a writing share x^* and a measure M^* of open topics such that: (i) $x^* = x^*(P^*)$ solves (5); (ii) newcomers are indifferent between joining any open topic and exploring: $w(P^*) = F(M^*)$; (iii) the allocation is stable under entry: $w'(P^*) < 0$; (iv) researchers add up: $\Phi(M^*) + M^* n(P^*) = N$.

Condition (iii) selects allocations that survive the adjustment $\dot{n}_j \propto w(P_j) - F(M)$. At a crossing where w rises, one more entrant raises the payoff of entering and draws in more. Topics in equilibrium need not have the same density a priori, but (ii) and (iii) leave a single admissible

density, so they do. We state results in the *large-topic limit* $\Phi(M)/N \to 0$. Here topics are large relative to the researcher time needed to open one, as they are in the data, where pioneers are a small share of authors. Appendix A gives the general expressions.

3 The payoff curve

Every result in the paper rests on the shape of v. The next theorem shows that it is single peaked for every learning and detection technology that satisfies Assumptions 1 to 3.

Assumption 4 (Interior reading and learning potential)**.** (a) Reading the first paper is worth its time: $b\,[\partial_R \ln Q(0,P) + \theta\Delta\varphi'(0)] > 1$ for all $P > 0$. (b) A thin literature is worth reading: for some $\varrho > 0$,

$$\nu_0(\varrho) \equiv \lim_{P\downarrow 0} \frac{Q(\varrho P, P) - 1}{P} - \frac{\varrho}{b} - \theta[\lambda - \Delta\varphi(\varrho)] > 0.$$

Part (b) compares, at the first papers of a topic, the gain from learning along the ray $R = \varrho P$ with the time spent reading and the overlap risk. In the example it reads $\sup_\varrho\{\gamma\beta\varrho/(1+\varrho) - \varrho/b - \theta[\lambda - \Delta(1 - e^{-\sigma\varrho})]\} > 0$.

Theorem 1 (The payoff curve)**.** *Under Assumptions 1 to 3:*

(i) *v is continuous on $\mathbb{R}_+$, $v(0) = 1$, $v(P) \geq e^{-\theta\lambda P} > 0$ and $\lim_{P\to\infty} v(P) = 0$.*

(ii) *$\ln v$ is concave on $(0,\infty)$. Under Assumption 4(a) it is strictly concave.*

(iii) *Under Assumption 4, there is a unique $\hat{P} \in (0,\infty)$ such that v is strictly increasing on $[0,\hat{P}]$ and strictly decreasing on $[\hat{P},\infty)$, with $\hat{v} \equiv v(\hat{P}) > 1$. For every $F \in (0,\hat{v})$ the equation $v(P) = F$ has exactly one root above $\hat{P}$.*

(iv) *(Envelope) v is differentiable on $(0,\infty)$ with*

$$\frac{v'(P)}{v(P)} = \underbrace{\partial_P \ln Q(R^*,P)}_{\text{more to learn from}} - \underbrace{\theta\big[\lambda - \Delta\big(\varphi(c^*) - c^*\varphi'(c^*)\big)\big]}_{\text{more to miss}}, \tag{7}$$

where $R^* = b\,r^*(P)$ and $c^* = R^*/P$.

(v) *The optimal reading share $r^*(P)$ is unique and nondecreasing in P. Papers per researcher, $a\,x^*(P)$, are nonincreasing in density.*

The proof (Appendix A) rests on one observation. $\ln g$ is *jointly* concave in reading time and density. $\ln(1-r)$ and $\ln Q$ are concave by assumption. The erosion exponent $-\theta\lambda P + \theta\Delta\, P\varphi(R/P)$ is linear plus the perspective of the concave function φ, and perspectives preserve concavity. Maximising a jointly concave function over one argument leaves a concave function of the other, so $\ln v$ is concave. A log-concave function that starts at one, rises, and vanishes at infinity is single peaked. Part (v) follows from Topkis's theorem: $\partial^2 \ln C/\partial R\,\partial P = -\theta\Delta\varphi''(c)\,c/P \geq 0$ and $\ln Q$ is supermodular, so reading and density are complements.

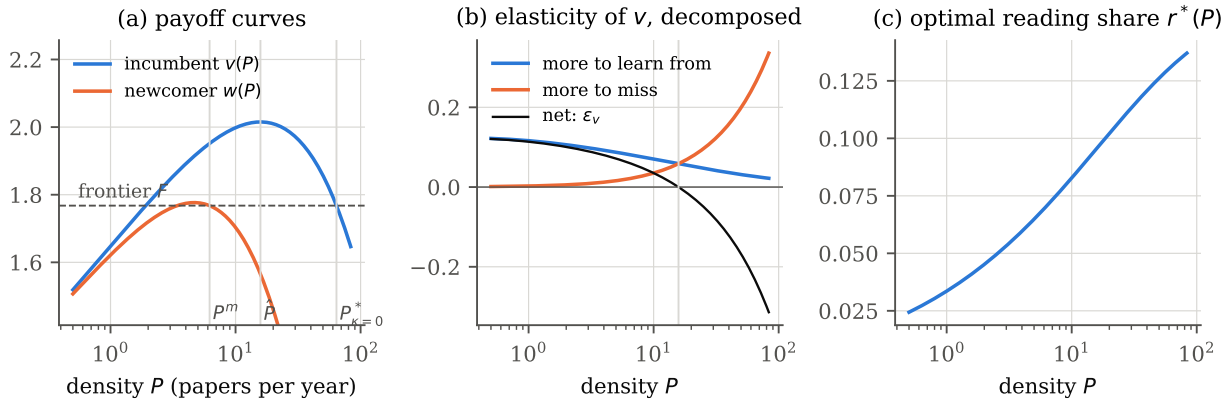


Figure 1: The payoff curve and its two forces. Left: $v(P)$ and the entrant's career payoff $w(P) = v(P)e^{-\kappa P}$ in the calibrated model, with the peak $\hat{P}$, the frictionless equilibrium (where $v = F$) and the mature density under catch-up costs (where $w = F$). Right: the decomposition (7) of v'/v into the learning and overlap terms, and the optimal reading share.

The theorem needs no parametric assumption, and its conditions are economic: no increasing returns to reading, and diminishing returns to search. Note what it does *not* need. Duplicates may keep most of their credit, since only $\lambda > \ell$ is required. The single peak comes from overlap that goes unread, not from an assumption that overlap is worthless.

Remark 2 (Where the hump comes from, and what the theorem rules out). The fall in v comes from *bounded absorption*. What a researcher can learn is bounded in density ($\bar{Q}(R) < \infty$) and what she can read is bounded by b, while the expected number of covering papers, θP, grows linearly with the literature. Any model with these two features has a falling payoff at high density. Joint concavity of $\ln Q$ is what makes the rise and the fall meet at a single peak. It rules out complementarities among the papers one reads, for example a literature that becomes useful only once a critical mass of results can be combined. With such complementarities $\ln v$ can have several local peaks, topics can have more than one stable size, and the comparative statics below apply locally around each falling-side crossing. We keep the concave case. It is the natural benchmark, and the arXiv returns profile (Fact 2) shows a single hump.

The sign of the spillover. Let one more paper appear in a topic. By the envelope theorem, the change in every other member's payoff is exactly $a\,v'(P)$. The researchers' own reallocation of time is second order.

Proposition 1 (Spillover sign)**.** *An additional paper in a topic of density P changes the per-period payoff of each incumbent by $a\,v'(P)$. The change is positive for $P < \hat{P}$ (more to learn from) and negative for $P > \hat{P}$ (more to miss).*

The sign of the spillover is therefore *derived*, not assumed. A paper helps its topic when the topic is thin and hurts it when the topic is dense. Overproduction is not built in. Whether science

is overcrowded depends on where equilibrium places topics on v. That is the subject of the next two sections.

4 Free topic choice without frictions

We first set $\kappa = 0$, so $w = v$. Newcomers move freely between topics and the frontier. This benchmark isolates the topic-choice margin.

Proposition 2 (Equilibrium and dissipation)**.** *Let Assumptions 1 to 4 hold, $\kappa = 0$, and suppose $\hat{v} > F(N/n(\hat{P}))$. Then a topic equilibrium exists and is unique. Every open topic sits on the falling side of the payoff curve, $P^* > \hat{P}$, where an additional paper lowers the payoff of every member (Proposition 1). Payoff per researcher equals the frontier payoff $a\,F(M^*) < a\,\hat{v}$. The gain from cumulative knowledge, $\hat{v} - F(M^*)$, is competed away.*

Entry continues past the peak because a newcomer compares the *average* payoff in a topic with the frontier. She ignores that her papers lower everyone else's payoff once the topic is past $\hat{P}$. Crossings on the rising side are unstable: one more entrant raises the payoff of entering. So topics fill past the peak, to the crossing on the falling side.

Remark 3 (Cities, clubs and topics). Propositions 2 and 4 have the structure of the theory of city sizes (Henderson, 1974) and of clubs (Buchanan, 1965; Scotchmer, 2002): a single-peaked average payoff in group size, free entry of new groups, stability only on the falling side, a planner at the peak of the average payoff, and an entry tax that decentralises it. Self-organised cities are too large for the same reason (Henderson and Becker, 2000). Topics are the cities, newcomers the migrants, and the frontier the hinterland where new cities are founded. Three things distinguish topics. The hump is not assumed but derived from the two jobs of reading (Theorem 1). The congestion is not in land or roads but in attention: the crowders are the readers, so whether an extra member helps or hurts is decided by where the topic sits on its payoff curve (Proposition 1). And the object the data measure, the duplication rate, is a function of density alone, which makes the invariance of Proposition 3 a testable restriction rather than a curiosity.

Volume opens new topics. The next result is the paper's anchor. It says how a larger community distributes itself.

Proposition 3 (Volume opens new topics)**.** *In the large-topic limit, equilibrium density depends on (N, a) only through the product Na, and*

$$\mathcal{E} \equiv \frac{\mathrm{d}\ln P^*}{\mathrm{d}\ln N} = \frac{\eta}{\eta\, D + |\varepsilon_w|}, \qquad D \equiv 1 - \varepsilon_x + \kappa P^* \geq 1, \tag{8}$$

where $\varepsilon_w = \mathrm{d}\ln w/\mathrm{d}\ln P < 0$ and $\varepsilon_x = \mathrm{d}\ln x^*/\mathrm{d}\ln P \leq 0$ are evaluated at P^*. New topics absorb a share $\mathrm{d}\ln M^*/\mathrm{d}\ln N = 1 - D\,\mathcal{E} = |\varepsilon_w|/(\eta D + |\varepsilon_w|)$ of growth, and papers per researcher move with elasticity $\varepsilon_x \mathcal{E} \leq 0$. If the frontier is perfectly elastic ($\eta = 0$), density, the overlap rate and the aware share do not depend on N, a or the cost of entering science.

Equation (8) holds for $\kappa \geq 0$; we use it again in Section 5. Three remarks.

First, the overlap rate is *pinned by entry, not by volume.* When the community grows, newcomers open new topics until existing ones are no more attractive than the frontier. The flagged overlap rate and the aware share are functions of P^* alone (Lemma 1), so they inherit its invariance. A flat overlap rate while submissions double (Fact 1) is what the model predicts. It is not evidence that crowding is harmless.

Second, a fixed field corresponds to $\eta \to \infty$. Then $\mathcal{E} \to 1/D$: growth thickens existing topics. The arXiv elasticities of 0.12–0.19 at broad radii and 0.035 (SE 0.118) at the closest-overlap radius, against fixed-field benchmarks of 0.53–0.81 (Fact 1), imply that most growth opens new ground. Inverting (8),

$$\eta = \frac{\mathcal{E}\,|\varepsilon_w|}{1 - \mathcal{E}\,D}, \tag{9}$$

so the frontier elasticity is identified by the volume elasticity and the slope of the entrant's payoff curve (Fig. 2, right).

Third, papers per author barely move when the community grows, because $\varepsilon_x \mathcal{E}$ is the product of two small numbers. Faster drafting, by contrast, raises papers per author one for one (Proposition 7). The two can be told apart only at the author level, because entrants change the composition of authors (Fact 5).

Remark 4 (Short-run and long-run elasticities). Proposition 3 compares steady states, while the arXiv elasticities are measured over 2018–2025, a period in which volume grew about 12% a year. In the model, mature topics adjust to a change in N at once, since the marginal newcomer is indifferent. The only wedge between the measured and the steady-state elasticity is composition. Faster growth means more young topics, which are thinner. If an area's growth rate rises by $\mathrm{d}g$ for T years, its age weights for topics younger than T scale by $e^{\mathrm{d}g(T-a)}$ and its log volume rises by $T\mathrm{d}g$, so

$$\mathcal{E}_{\text{measured}} = \mathcal{E} + \frac{\mathrm{Cov}_\omega\left(\ln P, (T-a)^+\right)}{T} + o(\mathrm{d}g), \tag{10}$$

where Cov_ω is the researcher-weighted covariance over the cross-section of topic ages a. The covariance is negative because young topics are thin, so the measured elasticity is a lower bound. In the calibrated life cycle the bias over a seven-year window is -0.073 (Section 9). The long-run elasticity is therefore at most about 0.073 above the measured one, which leaves the conclusion that volume mostly opens new ground intact. Steady growth at a constant rate does not bias the estimate at all, because area fixed effects absorb a constant age composition.

Remark 5 (Reading capacity per researcher or per field). Consider instead a fixed screening capacity B per field, so that coverage is B/P and researchers write full time. Topic choice still pins density, so Propositions 2 and 3 survive. But if topics cannot split, overlap rises with field volume, which the arXiv volume test rejects. Per-researcher capacity also keeps the time-allocation margin that separates writing tools from reading tools (Section 6). We adopt it throughout.

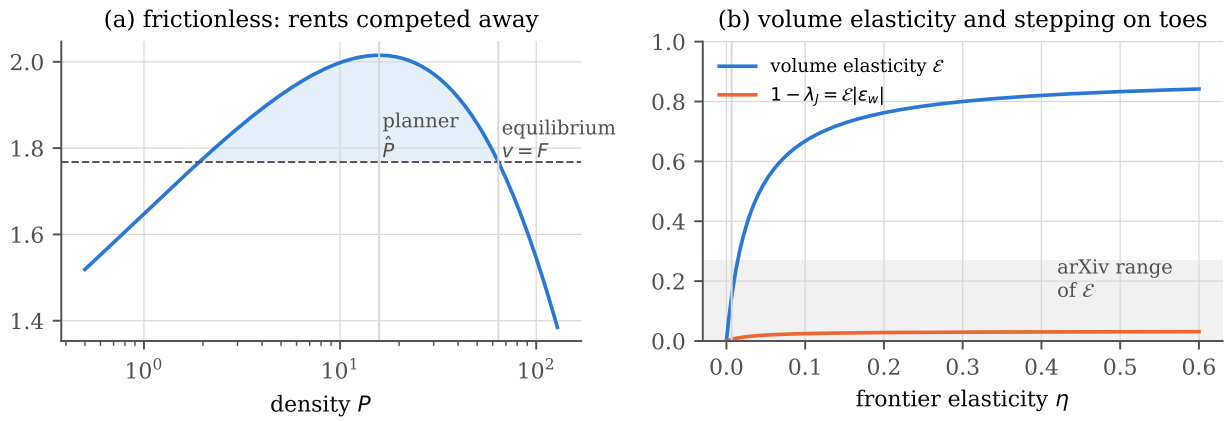


Figure 2: Left: without catch-up costs, topics fill to the density where $v = F$, and the shaded rent between the payoff curve and the frontier is competed away; the planner puts topics at the peak. Right: the volume elasticity $\mathcal{E}$ and the stepping-on-toes wedge $1 - \lambda_J = \mathcal{E}|\varepsilon_w|$ as functions of the frontier elasticity η, at the calibrated slope of w. The shaded band is the range of arXiv estimates of $\mathcal{E}$.

The planner. A planner who controls how many researchers join each topic, but not how they split their time, maximises $Y = a\,[f_0 M + M\,n\,w(P)]$ subject to $\Phi(M) + Mn = N$.

Proposition 4 (Overcrowding and the frontier subsidy)**.** *Let the conditions of Proposition 2 hold, with $\kappa \geq 0$.*

(i) The planner's density P^o satisfies

$$\frac{w(P^o) - F(M^o)}{w(P^o)} = |\varepsilon_w(P^o)|\,\frac{1 + n(P^o)\,\pi(M^o)}{D(P^o)}, \tag{11}$$

with $\hat{P}_w < P^o < P^*$ and $M^o > M^*$, where $\hat{P}_w$ is the peak of w. The planner opens more topics and puts fewer researchers in each. Overlap stays positive at the optimum. It is lower than in equilibrium whenever the credit lost to overlap rises with density, as it does in every calibration we report.

(ii) In the large-topic limit $P^o \to \hat{P}_w$. Topics sit at the peak of the entrant's payoff curve, which is the peak of v when $\kappa = 0$.

(iii) A frontier subsidy (equivalently, a tax on joining open topics) of $s^o = w(P^o) - F(M^o)$ per unit of writing speed decentralises the planner's allocation.

(iv) (Two wedges) A planner who also chooses x wants researchers to write less than $x^*(P)$ wherever $w'(P) < 0$ and more wherever $w'(P) > 0$. The writing wedge is proportional to $(\ln w)'(P)$. In the large-topic limit it vanishes at the optimal density, so controlling topic choice alone attains the first best.

There are two wedges, and they differ in importance. Newcomers choose topics on the average

payoff while the planner uses the marginal one: this is the *crowding wedge*. Inside each topic, researchers ignore that their writing adds to what colleagues must read: this is the *writing wedge*. Part (iv) shows the writing wedge is a symptom of the crowding wedge. Once topics are at their best density, researchers' private time allocation is efficient. Policies that work only inside topics are dissipated: writing caps, reading subsidies, better tools. With $\eta = 0$ the equilibrium payoff stays pinned at aF whatever happens inside topics. Policies that change the frontier are not dissipated: frontier grants, credit for first papers in a cluster, prizes for opening fields. The welfare criterion here is total private credit; Proposition 6 shows that the overcrowding result survives when part of that credit is a transfer.

5 Catch-up costs, rents and social value

The frictionless benchmark predicts equal returns across topics. The arXiv data reject this (Fact 2). The same researcher earns more citations in denser topics over most of the density range, and less in the densest ones. We now let $\kappa > 0$: newcomers must read a topic's stock before contributing.

5.1 Mature topics

Proposition 5 (Mature topics under catch-up costs)**.** *Let Assumptions 1 to 4 hold and* $\kappa > 0$.

(i) $w(P) = v(P)e^{-\kappa P}$ *is strictly log-concave, with a unique peak* $\hat{P}_w < \hat{P}$.

(ii) In a topic equilibrium the mature density P^m *solves* $w(P^m) = F(M)$ *on the falling side of* w. *Incumbents earn a* $v(P^m)$ *per period. Their log premium over the frontier equals the catch-up cost,* $\ln[v(P^m)/F(M)] = \kappa P^m = \rho T$, *where* $T = S/b_c$ *is catch-up time.*

(iii) Mature topics lie on the rising *side of* v *if and only if catch-up is costly relative to the rent at the peak:* $\kappa\hat{P} > \ln(\hat{v}/F)$.

(iv) Let topics differ in fertility ζ, *so that payoffs are* $\zeta v(P)$, *and let* $P^m(\zeta)$ *solve* $\zeta w(P) = F$. *Across mature topics, log returns rise with log density with slope exactly equal to the premium:*

$$\frac{\mathrm{d}\ln[\zeta\, v(P^m(\zeta))]}{\mathrm{d}\ln P^m(\zeta)} = \kappa P^m(\zeta) > 0.$$

Catch-up turns the dissipated rent into a protected one. Incumbents earn more than the frontier, and the premium exactly offsets what a newcomer gives up while reading the stock. More fertile topics attract more entrants, have larger stocks and so protect larger premia. This yields a positive returns–density gradient across topics, whether those topics lie left or right of $\hat{P}$. In the data the same gradient also reflects audience size, since a paper with more close neighbours has more potential citers. The measured gradient of 6.7% per standard deviation (Fact 2) is therefore an upper bound on the catch-up premium. We use it as such in Section 9.

5.2 Rising incumbent returns, falling entrant returns

The interesting object is the gap between what incumbents earn, v, and what a newcomer can expect over her career, $w = ve^{-\kappa P}$. Incumbents never pay the catch-up cost again. Newcomers and the planner do. The welfare statement follows almost by definition of stability, so we state it as a corollary and put the weight on what it implies for measurement.

Corollary 1 (Overcrowding with rising returns)**.** *In every stable topic equilibrium, one more member in a topic of density P changes total output of the other members by $n\,a\,w'(P)\,\mathrm{d}P/\mathrm{d}n < 0$. This counts both the spillover on incumbents (v') and the extra catch-up she imposes on future newcomers ($-\kappa v$). If $\kappa\hat{P} > \ln(\hat{v}/F)$, then $\hat{P}_w < P^m < \hat{P}$: incumbents' returns rise with density ($v'(P^m) > 0$) while the topic is too crowded ($w'(P^m) < 0$). The loss of output per researcher in the large-topic limit is $1 - F/\hat{w}$, where $\hat{w} = w(\hat{P}_w)$.*

The measured gradient is the incumbents' private return to density. A newcomer values a topic net of the reading she must do to join it, and so does the planner. Rising incumbent returns are the footprint of the catch-up rent, not of an untapped spillover. A natural reading of Fact 2 is that most researcher-years sit where more density raises returns, so most topics are too thin. The corollary says that this reading conflates v with w. Two kinds of topic can be too thin. The first is young topics that have not yet filled. The second, if the frontier is attractive enough, is newborn topics that never find followers (Remark 6).

Remark 6 (Coordination at the frontier). A lone pioneer earns less than a group, since v rises at first. Can new topics then stay too thin because nobody wants to be the first follower? With a continuum of topics and entry that responds to payoffs, the answer turns on a single inequality. If $F < v(P_0)$, the first follower of a newborn topic already beats the frontier, entry snowballs along the rising side, and no topic is stuck. If $F > v(P_0)$, newborn topics attract no followers, although a coordinated group would earn $\hat{v} > F$. Science then holds too many topics that are too small, and the planner would fix it by seeding followers. The calibrated model is in the first case (Section 9).

5.3 Social value and priority transfers

The welfare statements so far count private credit. Priority credit is partly a transfer (Merton, 1957; Hill and Stein, 2025a). When a later paper turns out to duplicate an earlier one, some of the credit it loses goes to the earlier paper, so part of the "loss" from overlap is redistribution. Conversely, a duplicate may be worth less to science than its residual citations suggest. We therefore separate social value from credit. Let the social value of a project be $Q\,C_s$, with

$$C_s(R, P) = \exp\{-\theta P[\lambda_s - \Delta_s \varphi(R/P)]\},$$

where λ_s is the share of a project's social value lost per unread covering paper and Δ_s the part of it that detection averts. Researchers still choose $r^*(P)$ to maximise credit, and social output

per unit of writing speed in a topic of density P is $s(P) = (1 - r^*)\, Q(br^*, P)\, C_s(br^*, P)$, with $s_w(P) = s(P)e^{-\kappa P}$. The catch-up cost is a real resource cost, reading time, and is not a transfer.

Proposition 6 (Social value)**.** *(i) One more member changes the social output of the others by* $n\, a\, s_w'(P)\, \mathrm{d}P/\mathrm{d}n$*, where*

$$\frac{s'(P)}{s(P)} = \frac{v'(P)}{v(P)} - \theta\big[(\lambda_s - \lambda) - (\Delta_s - \Delta)\psi(c^*)\big] + \theta b(\Delta_s - \Delta)\varphi'(c^*)\, r^{*\prime}(P),$$

with $\psi(c) = \varphi(c) - c\varphi'(c) \geq 0$*.*

(ii) (Uniform wedge.) *If* $\Delta_s = \Delta$*, then* $s(P) = v(P)e^{-\theta(\lambda_s - \lambda)P}$ *exactly. Social and private values then differ like a change in the catch-up cost, from* κ *to* $\kappa + \theta(\lambda_s - \lambda)$*. The equilibrium is overcrowded relative to the social optimum if and only if the share of overlap losses that are transfers,* $\tau \equiv (\lambda - \lambda_s)/\lambda$*, satisfies*

$$\tau < \frac{|\varepsilon_w(P^m)|}{\theta \lambda P^m}.$$

In particular overcrowding is stronger when duplicates are worth less to science than to their authors ($\lambda_s > \lambda$)*.*

(iii) (Proportional transfer.) *If a share* τ *of every overlap loss is a transfer,* $\ln C_s = (1-\tau)\ln C$*, the social optimum density maximises* $\ln v(P) - \tau \ln C^*(P) - \kappa P$*, where* $C^*(P) = C(br^*(P), P)$*.*

Part (ii) gives a sharp test. Overcrowding survives any degree of transfer as long as the slope of the entrant's payoff curve at the mature density exceeds the expected erosion intensity $\theta\lambda P^m$. In the calibrated model the slope is 0.037 against an erosion intensity of 0.025, so the critical transfer share is 1.5, above one. Under proportional transfers (part (iii)), the social optimum moves from 4.7 to 5.3 papers a year if all overlap losses are transfers, still below the mature density of 6.2. Transfers therefore shrink the welfare gain from optimal topic choice, from 0.5% to 0.1% of output per researcher, but they do not reverse its sign. The catch-up burden that newcomers impose on one another is a real resource cost, and on its own it keeps mature topics too dense. The transfer critique matters for magnitudes, and especially for the frictionless benchmark, where the dissipated rent is a difference in private payoffs; it does not matter for the direction of the inefficiency.

6 Faster writing scales science, better reading crowds it

New tools, artificial intelligence among them, enter the model through several parameters. Each leaves a different footprint, and the footprints of writing and reading tools point in opposite directions.

Proposition 7 (Footprints of new research tools)**.** *In the large-topic limit:*

(i) Faster drafting *($a \uparrow$) acts like a larger community. Density moves with elasticity $\mathcal{E}$ (zero if $\eta = 0$) and the number of topics with elasticity $1 - D\mathcal{E}$. Papers per researcher rise with elasticity $1 + \varepsilon_x \mathcal{E}$, and output per researcher with elasticity $1 - \mathcal{E}|\varepsilon_w|$.*

(ii) Cheaper entry into science *($c \downarrow$, with N set by $aF(M) = \omega + c$ and $\eta > 0$) raises N. Density rises with elasticity $\mathcal{E}$ with respect to N, papers per researcher move with elasticity $\varepsilon_x \mathcal{E} \approx 0$, and new topics absorb a share $1 - D\mathcal{E}$ of growth.*

(iii) Better reading, learning or search, or cheaper pivots *(b, β or $\sigma \uparrow$, $\kappa_p \downarrow$) raise v and w at every density. Density rises and the number of topics falls: science concentrates.*

(iv) Faster catch-up *($b_c \uparrow$, for example review articles or literature-summarising tools) raises w at every density, with the effects in (iii). It also lowers the overshooting threshold of Proposition 9, so more topics grow too fast.*

(v) Better search *($\sigma \uparrow$) raises the aware share A at any given reading coverage.*

Parts (iii) and (iv) follow from the envelope theorem. Each parameter raises $g(r, P)$ at every (r, P), hence raises v. A pointwise upward shift of a single-peaked curve moves its falling-side crossing with a fixed F to the right. The effect on the measured overlap rate is ambiguous in general: denser topics generate more overlap, but better detection removes more of it. Table 2 reports the calibrated magnitudes.

Proposition 8 (Reading improvements are dissipated)**.** *Let ω be any parameter that raises w at every density and leaves F unchanged. In the large-topic limit, $\mathrm{d}P^*/\mathrm{d}\omega > 0$ and $\mathrm{d}M^*/\mathrm{d}\omega < 0$, and*

$$\frac{\mathrm{d}\ln(Y/N)}{\mathrm{d}\omega} = -\eta \frac{\mathrm{d}\ln M^*}{\mathrm{d}\omega} = \eta \left| \frac{\mathrm{d}\ln M^*}{\mathrm{d}\omega} \right| \geq 0,$$

with equality if and only if $\eta = 0$. By contrast, $\mathrm{d}\ln(Y/N)/\mathrm{d}\ln a = 1 - \mathcal{E}|\varepsilon_w|$.

The asymmetry. Writing speed multiplies every payoff, including the frontier's. Newcomers spread out, so a faster pen raises output almost one for one and leaves duplication nearly unchanged in the long run. Reading speed improves only existing topics. Newcomers crowd in until those topics are no better than the frontier, so the improvement is competed away. Output per researcher equals the frontier payoff $aF(M^*)$, and a reading improvement raises it only because fewer topics are open and the remaining directions are more fertile, which is a gain of order η. A reading tool is worth subsidising, on this criterion, only if it also makes new topics easier to open, that is, if it raises F.

The asymmetry does not require a perfectly elastic frontier. Proposition 8 holds for any η, and the two elasticities can be compared directly. A proportional improvement in reading that raises $\ln w$ by $\mathrm{d}\omega$ at every density raises $\ln P^*$ by $\mathrm{d}\omega/(\eta D + |\varepsilon_w|)$ and output per researcher by $\eta D\,\mathrm{d}\omega/(\eta D + |\varepsilon_w|) = D\mathcal{E}\,\mathrm{d}\omega$. The same proportional improvement in writing raises output per researcher by $(1 - \mathcal{E}|\varepsilon_w|)\,\mathrm{d}\omega$. The ratio of the two gains is $D\mathcal{E}/(1 - \mathcal{E}|\varepsilon_w|)$, which is small exactly when the volume elasticity of density is small, as Fact 1 finds. The asymmetry is thus tied to an observable rather than to an assumption about F.

Table 2: The footprints of new research tools in the calibrated model

Channel	Shock	Density	Topics	Papers/ researcher	Output/ researcher	Flagged overlap	Aware share (pp)	g^* (pp)
Faster drafting	$a \times 1.3$	+3.7	+22.1	+26.1	+26.1	+4.9	−1.2	+0.0
Cheaper entry	$N \times 1.3$	+3.7	+22.1	−0.1	−0.1	+4.9	−1.2	+0.0
Faster reading	$b \times 1.3$	+30.7	−34.4	−0.2	+0.2	+31.7	−1.0	−4.1
Better learning	$\beta \times 1.3$	+47.7	−55.2	−1.4	+0.4	+63.4	−15.1	−11.4
Better search	$\sigma \times 1.3$	+1.5	−1.6	+0.0	+0.0	−10.1	+12.7	+0.5
Faster catch-up	$b_c \times 2$	+92.2	−97.5	−2.7	+0.7	+122.5	−27.2	−24.1
Cheaper pivots	$\kappa_p \div 2$	+7.8	−8.9	−0.3	+0.1	−26.9	−20.4	−0.4

Long-run stationary effects in log points (×100) of a 30% shock (catch-up speed doubled, pivot cost halved). Density is P^*; output per researcher is $aF(M^*)$. The flagged overlap rate is relative to its baseline level, which is a normalisation (Section 7). g^* is the myopic overshooting threshold of Proposition 9.

The proposition matches Evans (2008), who finds that online access narrowed the set of papers that get cited, and Hao et al. (2026), who find that AI tools expand individual scientists' impact but contract science's collective focus. Gains from access to frontier knowledge measured for individuals or topics (Iaria et al., 2018; Biasi and Moser, 2021) are real, but in equilibrium they are partly competed away at the level of science.

What the data can and cannot yet say. Sections 7 and 8 take the footprints to the arXiv record. At the author level, adopters of LLM-style writing raised their output by about 7% relative to comparable non-adopters, also in fields where AI is not the subject of research (Fact 5). The drafting channel is therefore present but so far small. The area-level evidence on pioneers and on overlap measured by meaning cannot separate channels: the first is confounded by changes in wording and the second is underpowered (Section 7). The reading and catch-up channels remain predictions. The model says that when literature-summarising tools become widespread, topics will get denser, fewer new ones will open, more fast-growing topics will overshoot, and output per researcher will barely rise. The arXiv rule that took most review articles out of computer science in November 2025 is a first shock to catch-up costs. Section 8 reports its first stage, and Section 11 states what the model predicts it will do.

7 Evidence I: the arXiv record

This section documents five facts that discipline the model. The analysis frame holds 1.49 million arXiv papers posted in 2018–2025 (1.65 million through mid-2026), classified into 150 text-based fine areas. Volume rose by 122% over 2018–2025. It is complemented by a sample of 25,400 full-text PDFs for citation behaviour, by citation counts and reference lists, and, for the validation of awareness, by curated INSPIRE-HEP reference lists. A paper's *overlap flag* is set when an earlier paper by another group, posted 3 to 24 months before, lies above its area's 99th percentile of 2018–19 nearest-neighbour TF-IDF similarity. The *level* of the flagged rate, about 1% in the base period,

is a normalisation; its *changes* carry the information. Text-based similarity follows Hoberg and Phillips (2016) and Kelly et al. (2021). TF-IDF and sentence-embedding flags agree on only about a fifth of flagged papers, so we report which measure each fact uses and do not lean on either alone. Table 3 collects the numbers. Standard errors are clustered by author unless noted. Clustering by area would be more conservative and is part of the replication package.

Fact 1. *Volume opens new ground.* We regress the log local-neighbour hazard, the rate at which a paper's similarity radius is populated by earlier papers, on the log of the area's prior two-year volume with area and year effects. From the 95th to the 25th percentile radius, the elasticity is 0.12–0.19. If an area's papers kept landing in the same idea space, randomly thinning each paper's observed neighbour list implies 0.53–0.81 at the same radii. At the 99th percentile radius, the radius of the overlap flag, the estimate is 0.035 with a standard error of 0.118 (95% interval −0.20 to 0.27). In the all-arXiv time series, volume rose 122% from 2018 to 2025 while the overlap hazard rose about 13% (elasticity 0.19–0.27). The model's topic is a cell of fixed size in idea space (Remark 1), so its density is the close-radius density. We therefore use $\mathcal{E} = 0.15$ as the baseline and report results over $[0, 0.27]$. Broad radii fill faster than close ones, as new topics are packed into an area's idea space (prediction P7 in Section 11).

Fact 2. *Returns are hump-shaped in topic density.* Comparing the same researcher across years spent in denser or sparser topics, with author and year effects (923,000 author-years), one standard deviation of density goes with 6.7% more fractional citations per author-year (SE 0.15 percentage points, clustered by author). Relative to the middle fifth of density, returns are −11% in the sparsest fifth, +2% in the fourth fifth and −2% in the densest. The densest researcher-years thus earn less than the middle. A quadratic puts the peak about one standard deviation above mean density, with 12% of researcher-years beyond it. The shape survives using the third-closest paper and restricting to researchers active three or more years. Two caveats matter for interpretation. First, citations reward audience. A paper with more close neighbours has more potential citers, which by itself produces a rising gradient, so the rising arm is an upper bound on the returns to density. Second, papers per author-year rise slightly with density within author, +0.5% per standard deviation (SE 0.1). Theorem 1(v) predicts a fall. The discrepancy is small, but it is a failed prediction and we report it. Larger teams in denser topics are one candidate explanation.

Fact 3. *Duplicates lose some credit, aware follow-ons gain.* Among 2018–2021 papers with at least five references, relative to papers in the same area-year with the same topic density, *unaware* overlaps (which do not cite their nearest earlier paper) receive 19% fewer citations (SE 2). *Aware* overlaps (which cite it) receive 29% more (SE 5). The penalty is close to the 21% citation penalty for scooped teams in structural biology estimated by Hill and Stein (2025a) with a different design and field. The two estimates validate each other.

Fact 4. *About 60% of close overlap is aware.* In the PDF sample, the raw share of flagged pairs in

Table 3: The arXiv facts

Fact	Statistic	Value
1	Elasticity of local density to area volume, 95th–25th pct. radius	0.12–0.19
	at the overlap radius (99th pct.)	0.035 (0.118)
	fixed-field benchmark at the same radii	0.53–0.81
	all-arXiv time series, 2018–2025 (volume +122%)	0.19–0.27
2	Citations per author-year per SD of density (author FE)	+6.7% (0.15 pp)
	By fifth, relative to the middle: sparsest / 4th / densest	−11% / +2% / −2%
	Peak of quadratic; share of researcher-years beyond it	≈ +1 SD; 12%
	Papers per author-year per SD of density	+0.5% (0.1)
3	Citations: unaware overlap / aware overlap	−19% (2) / +29% (5)
4	Aware share of close overlap: raw / recall-corrected	35–42% / ≈ 60%
	INSPIRE-HEP, 90th / 75th pct. similarity	61% / 50%
5	Author-level output of LLM-style adopters, 2024–26	≈7%

Notes: arXiv 2018–2025, 150 areas. Standard errors in parentheses. Facts 1–4 discipline the calibration of Section 9. Fact 5 bears on the drafting channel of Proposition 7.

which the later paper cites the earlier one is 35–42% in every year from 2019 to 2026. PDF matching recovers 63% of citations, so the recall-corrected share is about 60%. Recall is not constant: arXiv identifiers recovered per PDF rose from 5.0 in 2019 to 8.8 in 2026, and the raw aware share rose by 3.8 percentage points (SE 1.3) after 2022. Only the comparison across areas with high and low exposure to large language models is flat (−0.5 pp per standard deviation, SE 1.4). An independent check comes from high-energy physics, where INSPIRE-HEP curates complete reference lists. Among later papers by other authors that overlap an earlier paper at the area's 90th percentile of nearest-neighbour similarity, 61% cite it (50% at the 75th percentile). We calibrate to 60%.

Fact 5. *Growth came mostly from new authors, and drafting tools raised output per adopter only modestly.* At the area level, papers per unique author identifier stayed flat while the number of authors grew. An area-level ratio cannot rule out a drafting effect, because entrants change the composition of authors. Author identifiers are also unreliable in 2025, when many were reissued. We therefore build an author-level panel with name-keyed submitter identities (Section 8). Submitters whose 2023 abstracts adopt the vocabulary that spread with large language models (Liang et al., 2024; Kobak et al., 2025) raised their output in 2024–2026 by about 7% relative to submitters who did not, and by a similar amount in fields where AI is not the subject. Kusumegi et al. (2025) report 36% more papers for LLM adopters. Their timing rule, which dates adoption by the first flagged paper, mechanically generates positive dynamics (Renault et al., 2026). Classifying adoption before the outcome window, as we do, avoids this.

What we no longer claim. An earlier version of this evidence reported that areas with high LLM uptake produced more pioneer papers, and that overlap measured by meaning did not fall.

Neither survives scrutiny. The pioneer measure uses the same TF-IDF neighbours that changes in wording distort. In high-uptake areas, rare words in abstracts rose from 25.4 to 33.0 per abstract and hyphenated words from 6.9 to 10.5 over 2025–2026. Without detrending against a pre-existing decline, the pioneer estimate is −0.8 pp (SE 0.2), which points to concentration rather than dispersion. The embedding-based overlap estimate, −0.19 pp (SE 0.24) on a base near 1%, has a minimum detectable effect at 80% power of about 0.67 pp, roughly two thirds of the base. Most model channels move flagged overlap by 7–30% (Table 2), so the test cannot discriminate. Finally, 13 of the 15 highest-uptake areas study AI or computer vision, so area exposure confounds AI as a tool with AI as a subject. The author-level design in Fact 5 addresses this by restricting to fields where AI is not the subject.

Mapping to the model. Fact 1 is the volume elasticity $\mathcal{E}$ of Proposition 3. Fact 2 rejects the equal returns of the frictionless model and motivates the catch-up cost of Section 5. Fact 3 disciplines the credit parameters λ and Δ/λ (Lemma 1), and Fact 4 disciplines search precision σ through the aware share (4). None of these facts is causal. The next section turns to designs that are.

8 Evidence II: arXiv's rules as experiments

The facts above are correlations. This section uses arXiv's own rules, which create variation in how much a paper is read, in the cost of entering, in the cost of catching up on a literature and in how much one person may write. The data are the arXiv metadata snapshot of 18 October 2026 (Cornell University, 2026): 2.2 million papers with new-style identifiers submitted from 2015 to 15 October 2026, with the exact time of each submission, the submitter's name, the author list and the abstract. Appendix D documents every construction. Tables 4 and 5 report the estimates.

Reconstructing the daily listings. arXiv announces new papers once a day. Papers submitted before the daily cutoff appear in that evening's listing, and later papers wait for the next one. Identifiers are assigned at announcement, in listing order. Within an announcement batch, the rank correlation between identifier and submission time is 0.98 for papers that were not held for moderation (median across batches). Six percent of papers are held and announced later. The submission-time distribution reveals the cutoff. It was 16:00 Eastern Time until the end of 2016 and has been 14:00 since January 2017. On an average weekday about 15 papers are submitted in the first ten seconds after 14:00:00: authors race for the top of the next listing (Haque and Ginsparg, 2009). We place every paper in its announcement batch and rank it within its primary category's list of new submissions.

8.1 Does reading protect? Listing position as a readership shock

Lemma 1 gives sharp predictions for a shock that makes a paper i more visible to the researchers who might later cover it. If visibility raises the probability φ that a later covering project detects i, unaware duplicates of i fall by $\theta\,\mathrm{d}\varphi$, aware follow-ons that cite i rise by $q\theta\,\mathrm{d}\varphi$, and the ratio of the two effects identifies q, the share of detections that proceed. We test these predictions with two designs.

Design A: the daily cutoff. A paper submitted just after 14:00 opens the next listing, while one submitted just before closes the current one. The first stage is sharp (Table 4, panel A; Fig. 3). Within an hour of the cutoff, excluding a five-minute donut around the bunching, crossing the cutoff raises the probability of being among the first five papers by 0.69, from a base of 0.30. The comparison it delivers, however, is first against *last*. The end of a listing is also highly visible, a "reverse visibility" effect documented for arXiv by Haque and Ginsparg (2010) and for NBER working papers by Feenberg et al. (2017). Submitters on the two sides also differ: those just after the cutoff have fewer prior submissions (-0.051, SE 0.017). Top-of-listing papers receive *fewer* citations than end-of-listing papers: -0.137 log points (SE 0.060) in high-energy physics, where INSPIRE-HEP gives complete citation counts, and -0.175 (SE 0.032) across all fields in the 43,097 papers with Semantic Scholar counts. The cutoff therefore does not isolate visibility, and we do not use it to test detection.

Design B: rank within the listing. A cleaner source of variation is how many other papers happen to be submitted earlier on the same day. A paper's position is one plus the number of papers in its category submitted since the batch opened. We hold fixed the submission time (in 15-minute bins, separately for the weekend batch), the category-year, the announcement date and the size of the listing. Position then varies only with the timing of other people's submissions. We use all 87,694 papers in the nine categories covered by INSPIRE-HEP in 2017–2022, excluding the first 15 minutes after the cutoff. The residual standard deviation of log position is 0.45. Predetermined characteristics are balanced: number of authors and the submitter's prior submissions are unrelated to residual position. Abstracts are slightly shorter lower in the list (1.6 words of 151 per log point, SE 0.6).

The readership effect is small. One log point lower in the listing, roughly moving from second to fifth place, reduces citations by 0.023 log points (SE 0.016; 0.031 without date fixed effects). Position bins show no premium for the very first slots once dates are held fixed. Later overlapping work responds even less. We count later papers in the same category, within 24 months, by authors who do not overlap with the focal paper's, whose TF-IDF similarity exceeds the category's median (75th percentile) of nearest-neighbour similarity. With INSPIRE's curated reference lists we split them into those that cite the focal paper (aware) and those that do not (unaware). One log point lower in the listing changes unaware overlap at the 75th percentile threshold by -0.0049 (SE 0.0038) on a mean of 0.068, and aware overlap by -0.0001 (SE 0.0041). The 95% interval rules out an

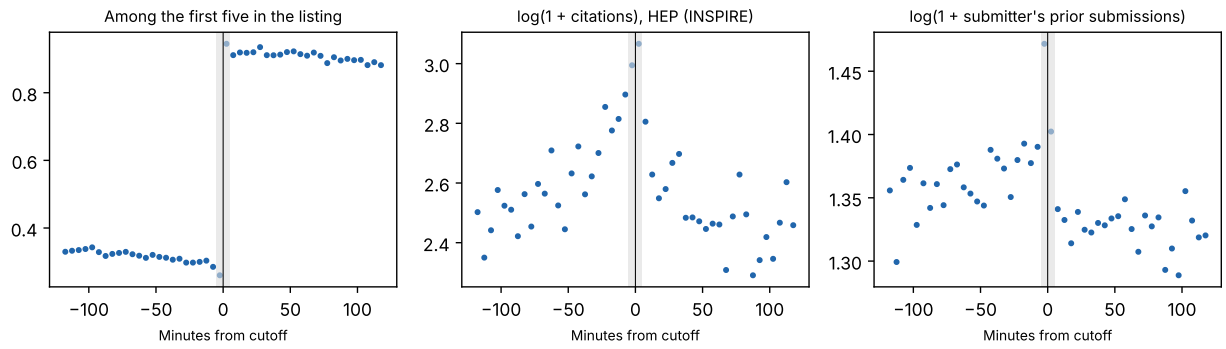


Figure 3: The daily cutoff. Binned means, net of category-year means, by minutes from the 14:00 ET cutoff (weekday submissions 2017–2022; shaded: five-minute donut). Left: probability of being among the first five papers of the category's listing. Middle: log citations in high-energy physics (INSPIRE-HEP). Right: the submitter's prior submissions, a predetermined characteristic that jumps at the cutoff. Papers just after the cutoff open the next listing; papers just before close the current one.

increase in unaware overlap of more than 3.9% per log point. The minimum detectable effect at 80% power is 16% of the mean. Self-follow-ups by the focal paper's own authors fall with position (−0.048, SE 0.015). This suggests that attention encourages authors to continue a line of work, or that the design leaves some day-level confounding; either way it does not affect the comparison with other authors' work.

What this teaches. Passive visibility moves casual readership a little and does not move the detection of overlapping work. In the model, detection depends on the coverage R/P that a later researcher chooses and on search precision σ, not on how prominently the earlier paper was announced. The result says that the protective job of reading is done by targeted search when a project is planned or written up, not by browsing daily listings. This matches the high calibrated search precision needed for a 60% aware share (Fact 4). The right causal test of φ is therefore a shock to the cost of search. Section 11 sets out one: the blocking of Google in mainland China from 2014 (Zheng and Wang, 2020). The model predicts that it lowered the aware share and raised unaware duplication in work by China-based authors, relative to Hong Kong-based and overseas Chinese authors in the same topics.

8.2 Entry cost: the endorsement rule

New arXiv submitters need an endorsement. Until late 2025, an e-mail address at a recognised institution sufficed. On 10 December 2025 arXiv stopped accepting it alone in Mathematics (arXiv, 2025b), and on 21 January 2026 in all sections (arXiv, 2026a). New authors now need prior authorship of an accepted paper in the section or a personal endorsement. We measure the share of papers whose submitter has never submitted before. We compare each week with the same week of the previous academic year, which removes seasonality, and Mathematics with other sections,

Table 4: Listing position, readership and later overlapping work

Outcome	Estimate	(SE)	Mean	N
A. Discontinuity at the daily cutoff: submitted just after (top of next listing) vs. just before (end of current listing)				
Position in listing (0 = top, 1 = bottom)	−0.838	(0.004)	0.884	77,051
Among the first five papers	+0.688	(0.006)	0.300	77,051
Number of authors (balance)	−0.178	(0.454)	5.650	77,051
Abstract length, words (balance)	−0.832	(0.917)	153.785	77,051
log(1 + submitter's prior submissions) (balance)	−0.051	(0.017)	1.363	77,050
log(1 + citations), HEP (INSPIRE)	−0.137	(0.060)	2.682	11,483
log(1 + citations), all fields (Semantic Scholar)	−0.175	(0.032)	2.522	43,097
B. Listing rank in high-energy physics: effect of log position, holding submission time fixed				
log(1 + citations)	−0.023	(0.016)	2.309	87,679
Citation percentile in category-year	−0.005	(0.003)	0.480	87,679
Later overlapping papers by others, τ_{50}	−0.021	(0.018)	0.667	87,694
of which cite the focal paper (aware)	−0.003	(0.008)	0.211	87,694
of which do not (unaware)	−0.018	(0.014)	0.454	87,694
Later overlapping papers by others, τ_{75}	−0.005	(0.006)	0.130	87,694
aware	−0.000	(0.004)	0.062	87,694
unaware	−0.005	(0.004)	0.068	87,694
Later overlapping papers by the same authors, τ_{50}	−0.048	(0.015)	0.687	87,694

Notes: Panel A: local linear regression discontinuity at the 14:00 ET cutoff, bandwidth 60 minutes, triangular kernel, donut of 5 minutes, separate slopes, category-year and weekday fixed effects; standard errors clustered by submission day; "Mean" is the mean just before the cutoff. Panel B: all papers in hep-th, hep-ph, hep-ex, hep-lat, gr-qc, nucl-th, nucl-ex, astro-ph.CO and astro-ph.HE announced in their regular batch, 2017–2022, submitted at least 15 minutes after the batch opened; regressors are log listing position and log list size; fixed effects for category-year, submission-time bin (15 minutes; hourly for the weekend batch) and announcement date; standard errors clustered by announcement date. Overlap thresholds τ_q are percentiles of the category's nearest-neighbour similarity distribution. Citations and reference lists from INSPIRE-HEP (excluding self-citations) unless noted.

which exploits the staggered timing.

The first stage is large and well timed (Table 5, panel A; Fig. 4). Net of the year before, the first-time submitter share fell by 2.4 percentage points in Mathematics (SE 0.3; about 17% of its base) and by 3.4 points in other sections (SE 0.3; about 15%). In the six weeks when only Mathematics was treated, Mathematics fell 1.7 points more than the other sections (SE 0.4). Submitters with no prior arXiv authorship, the group the rule targets, fell by about 17% outside Mathematics. In the model, this is a rise in the cost of entering science. Proposition 7(ii) predicts that it reduces the number of topics roughly in proportion to the entrants it removes, while leaving density, the overlap rate and the aware share nearly unchanged, because $\mathcal{E}$ is small. Those outcomes need a year of post-period data; Section 11 states them in advance.

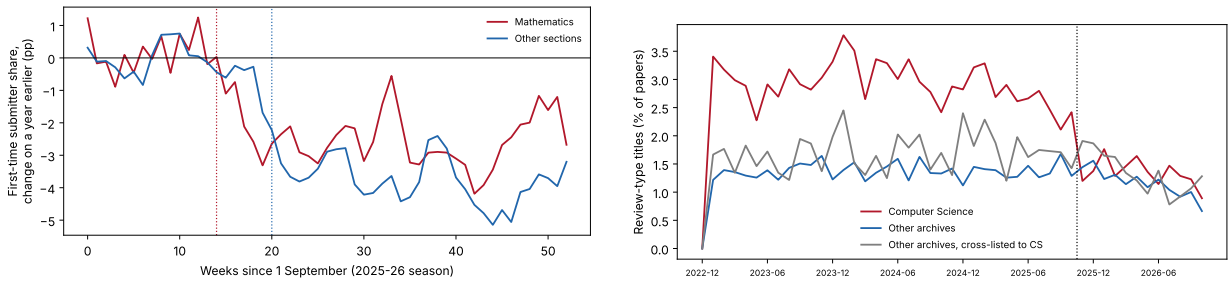
8.3 Catch-up cost: review articles in computer science

Review articles summarise a literature and lower the cost of catching up on it (McMahan and McFarland, 2021). On 31 October 2025, arXiv began to require that review and position papers in computer science be accepted after peer review before posting (arXiv, 2025a). The share of computer-science papers with review-type titles fell from 2.7% to 1.3%, from about 290 to 187 a month. Relative to other archives the fall is 1.32 percentage points (SE 0.10), and it survives a CS-specific trend. Reviews did not migrate to other primary categories cross-listed to computer science (Table 5, panel B). Within computer science, categories that had relied more on reviews lost more of them. A category with a 1 point higher pre-period review share lost 0.41 points more (SE 0.10), against an insignificant placebo a year earlier.

In the model, fewer reviews mean slower catch-up, a higher κ and therefore lower mature density, more topics and fewer overshooting topics (Proposition 7(iv)). The short-run implication is less entry into existing topics where reviews mattered. The share of papers with a first-time author fell more in review-reliant categories: -0.70 per unit of exposure (SE 0.26). The same regression a year earlier gives -0.32 (SE 0.17), so part of the gradient predates the rule. We read the reduced form as suggestive. The topic-level predictions are stated in Section 11.

8.4 A cap on writing: two submissions per person per month

On 1 October 2026, arXiv limited every submitter to two new submissions per calendar month, without prior announcement and with no exemptions (arXiv, 2026b). The cap applies to the submitting account, not to the authors. Because shared names belong to different people, we measure excess submissions among names whose submissions in a half-month fall in a single archive. In the first half of each month of 2026 before the cap, 1.66% of papers were a submitter's third or later submission; in October the share was 0.38%, a residual consistent with shared names and proceedings accounts. The cap therefore removed about 1.3% of papers mechanically. Volume did not fall. Papers per weekday rose 0.220 log points from the first half of September to the first half of October 2026, against 0.175 on average in 2024–2025. Categories more exposed to heavy submitters did not grow less (Table 5, panel C). Per-submitter caps bind on a sliver of output and



(a) Endorsement rule

(b) CS review policy

Figure 4: Two arXiv rule changes. (a) First-time submitter share, change on the same week a year earlier (three-week moving average), Mathematics and other sections; dotted lines mark 10 December 2025 and 21 January 2026. (b) Share of papers with review-type titles by archive; the dotted line marks November 2025.

can be met by letting a co-author submit. In the model's terms, the cap is not a shock to writing speed, and the model predicts that a binding one, being a policy inside topics, would be largely dissipated by entry.

8.5 Drafting speed: author-level adoption of LLM-style writing

Proposition 7(i) predicts that faster drafting raises papers per researcher one for one. To measure drafting at the author level, we follow the excess-vocabulary literature (Liang et al., 2024; Kobak et al., 2025). We flag abstracts that use words whose frequency jumped after ChatGPT ("delve", "showcase", "underscore", "intricate", "pivotal", and similar). Their share rose from 3% of abstracts in 2022 to 11% in 2024. Among 68,467 submitters active in 2021–2022 and 2023, the 4,488 who used such words in at least half of their 2023 abstracts are adopters. Submitters who never used them are the comparison group. Classification uses 2023 papers only, so the outcome window, 2024–2026, is not conditioned on it. With submitter and field-by-half-year fixed effects, adopters' log(1 + papers per half-year) rose by 0.022 (SE 0.005), about 7% more papers, and by 0.024 (SE 0.007) in fields where AI is not the subject (Fig. 8 in Appendix D). Pre-period coefficients drift by up to 0.039, so part of this is a pre-existing trend. The drafting channel is present but small: adopters have so far used language models to write differently more than to write more.

8.6 What the shocks say about the model

The four designs do not test every proposition, and two of them are first stages whose outcomes are still accruing. Together they establish three things. The entry margin responds to its cost, which is what makes density respond little to volume. The supply of review articles, a lever on catch-up costs, responds to policy, and the first reduced form on entry has the sign the model predicts. And the detection of overlapping work does not run through passive visibility. Reading protects through deliberate search, which is why better search tools, not more prominent announcements, are the

Table 5: arXiv rule changes as shocks to entry, catch-up and writing

Design and outcome	Estimate	(SE)	Pre-period mean
A. Endorsement tightening (entry cost): Mathematics 10 Dec 2025, all sections 21 Jan 2026			
First-time submitter share, Mathematics, vs. year before	−0.024	(0.003)	0.141
First-time submitter share, other sections, vs. year before	−0.034	(0.003)	0.222
First-time submitter share, Mathematics vs. others, 10 Dec–20 Jan	−0.017	(0.004)	
Submitters with no prior arXiv authorship, Mathematics	−0.009	(0.003)	0.043
Submitters with no prior arXiv authorship, other sections	−0.009	(0.002)	0.054
log number of first-time submitters, other sections	−0.154	(0.035)	
B. Peer review required for CS review and position papers (catch-up cost), 31 Oct 2025			
Review-type share of papers, CS vs. other archives	−0.013	(0.001)	0.027
with a CS-specific linear trend	−0.010	(0.002)	
leakage: non-CS primary cross-listed to CS	−0.002	(0.001)	
Within CS: change in review share on pre-period review share	−0.41	(0.10)	0.027
placebo, one year earlier	−0.13	(0.09)	
Within CS: change in first-time-author share on review share	−0.70	(0.26)	
placebo, one year earlier	−0.32	(0.17)	
C. Two submissions per submitter per month (writing cap), 1 Oct 2026			
Third-or-later submissions in days 1–15, single-field names: Jan–Sep 2026	1.66%		
October 2026 (residual: shared names and proceedings accounts)	0.38%		
log papers per weekday, Oct 1–15 vs. Sep 1–15: 2026	+0.220		
same change in 2024 and 2025	+0.173, +0.178		
Category dose-response: change on exposure (vs. 2025)	+0.03	(2.95)	0.016
D. Author-level adoption of LLM-style writing (drafting speed)			
log(1 + papers per half-year), adopters vs. others, 2024–26, all fields	+0.022	(0.005)	0.48
fields where AI is not the subject (no cs, stat, eess)	+0.024	(0.007)	0.52

Notes: Panel A: weekly cells by announcement week, September 2024–October 2026; each week is differenced against the same week a year earlier; Newey–West standard errors (four lags). Panel B: monthly cells, 2023–2026, archive and month fixed effects, robust standard errors; dose-response across 29 CS categories with at least 300 papers a year, weighted by size, Nov–Oct years. Panel C: days 1–15 of each month, papers announced in their regular batch; exposure is the category's Jan–Sep 2026 share of excess submissions. Panel D: submitter and field-by-half-year fixed effects, standard errors clustered by submitter. Identities are name keys throughout (Appendix D).

channel through which AI could move the aware share. A per-person writing cap, finally, is not a lever on volume at all.

9 Quantification

This section puts numbers on the mechanisms. The aim is not to fit every moment of the arXiv record. It is to check that one parameterisation of the model reproduces the main facts with plausible primitives, and to read off its implications for rents, overcrowding, new research tools and growth.

9.1 Strategy

We use the functional forms of Section 2: harmonic learning (2) and exponential detection. Time is measured in years. Table 6 lists the parameters. Some are set externally. The unit of density is fixed by $b = 100$ papers read per year at full-time reading. Careers leave a topic at $\rho = 0.10$ (ten-year tenure), the stock's relevance decays at $\delta = 0.25$ (a half-life of about three years), topics become obsolete at $\xi = 0.04$, and a newborn topic produces $P_0 = 2$ papers a year. The credit block comes from Fact 3 through Lemma 1. With exponential follow-on credit and a pivot that costs 30% of a project, the +29% premium of aware follow-ons and the −19% penalty of unaware duplicates imply that 31% of detected overlaps proceed and that detection averts $\Delta/\lambda = 37\%$ of erosion.

The remaining parameters are calibrated jointly: learning (β, γ), erosion $(\theta\lambda)$, search precision (σ), the frontier payoff F, catch-up speed b_c, and the distributions of diffusion rates and fertility. The targets are the corrected returns profile of Fact 2: the gradient per standard deviation of density, the returns of the sparsest, fourth and densest fifths relative to the middle one (−11%, +2%, −2%), and the 12% of researcher-years beyond the peak. We add the recall-corrected aware share of 60% at mature density (Fact 4), which disciplines σ. A soft target puts the incumbents' log premium $\kappa P^m = \rho T$ at about 0.15, a year and a half of catch-up per ten-year tenure. Since the measured gradient also reflects audience, this is generous to the catch-up channel. We simulate the stationary cross-section of topic life cycles and run the arXiv regressions on it. As in the data, density is measured with error. Only 58% of a paper's nearest neighbours fall in its own fine area. We add classical noise with a standard deviation of 0.3 in log density, which implies a reliability of about 0.8. Researchers who move between topics supply the identifying variation in an author-fixed-effects design, and the model has no ability differences. The baseline analogue is therefore the cross-section of researcher-years. The within-topic analogue, which uses only life-cycle variation, is a robustness check. The search is a quasi-random scan of 2,048 points followed by Nelder–Mead from the best points, with the winner selected on a fine simulation grid (Appendix C). The frontier elasticity η is then set from (9) with $\mathcal{E} = 0.15$ (Fact 1).

Table 6: Parameters

Parameter	Meaning	Value	Source
Set externally			
b	papers read per year at full-time reading	100	normalisation (unit of P)
λ	credit lost by an unaware duplicate	0.19	Fact 3 (-19%)
κ_p	credit lost by pivoting	0.30	assumption
Δ/λ, q	detection value; share proceeding	0.37, 0.31	Fact 3 ($+29\%$), Lemma 1
δ	decay of the stock's relevance	0.25	half-life ≈ 2.8 years
ρ	rate of leaving a topic	0.10	ten-year topic tenure
ξ	topic obsolescence	0.04	25-year topic life
P_0	flow of a newborn topic	2	a few papers per year
Calibrated (Table 7)			
β, γ	learning efficiency, curvature	39.01, 0.156	hump in returns
$\theta\lambda$	erosion per earlier paper	0.0041	fall at the top
F	frontier payoff	1.768	sparsest fifth
b_c	catch-up reading speed	25	premium, share beyond peak
σ	search precision	1.58	recall-corrected aware share
g_{med}, s_g	diffusion rate: median, log-sd	0.182, 0.29	share beyond peak, fifths
s_ζ	log-sd of topic fertility	0.052	gradient across topics
η	frontier elasticity	0.0067	$\mathcal{E} = 0.15$ (Fact 1), (9)

9.2 Fit

Table 7 and Fig. 5 compare model and data. The calibrated model reproduces the gradient (6.4% against 6.7%), the rise from the sparsest fifth (-11.0% against -11%), the share of researcher-years beyond the peak of the payoff curve (12% against 12%), the aware share at mature density (60% against 60%) and, without targeting it, the location of the peak (0.73 standard deviations above mean density, against about one). The incumbents' premium falls short of its soft target, which is generous to the catch-up channel because the measured gradient also reflects audience.

The model does not reproduce the top of the profile. In the corrected data the densest fifth earns *less* than the middle (-2%). In the model the fourth and densest fifths earn $+1.4\%$ and $+2.5\%$. The source of the miss is composition. The densest researcher-years in the model include the most fertile mature topics, whose incumbents earn high returns, alongside the overshooting topics that pull returns down. Erosion in mature topics is small, about 1.7% of credit, so the payoff curve falls only gently beyond its peak. Searches that allow faster diffusion find corner solutions in which almost every topic overshoots, without a lower loss (Appendix C). The fall at the top points to crowding costs in dense topics beyond duplication and catch-up, or to audience effects that flatter the middle of the distribution. We regard it as the model's main quantitative failure. 48% of researcher-years are in mature topics, 25% in topics consolidating after a growth spurt, and 27% in growing topics.

9.3 What the calibrated model says

Table 8 collects the implications.

Table 7: Fit to the arXiv facts

Moment	arXiv	Baseline	Within-topic
Gradient per SD of density	0.067	0.064	0.057
Sparsest fifth vs. middle	−0.110	−0.110	−0.115
Fourth fifth vs. middle	0.020	0.014	0.010
Densest fifth vs. middle	−0.020	0.025	0.012
Researcher-years beyond the peak	0.120	0.123	0.158
Incumbents' log premium κP^m	0.150	0.099	0.184
Peak of v (SD above mean)†	1.00	0.73	0.87
Quadratic: linear term†	0.094	0.047	0.048
Quadratic: squared term†	−0.047	−0.009	−0.014
Aware share of flagged overlap	0.60	0.60	0.60

† Not targeted. Baseline: cross-section of researcher-years (movers across topics). Within-topic: topic fixed effects, recalibrated. Density is the researcher-weighted standardised log flow, measured with noise as in the data.

Overcrowding and rents. The payoff curve peaks at $\hat{P} = 16$ papers a year. Mature topics sit below the peak, at $P^m = 6.2$, and the entrant's payoff curve peaks at $\hat{P}_w = 4.7$. Incumbents earn a premium of 10% over the frontier, the reward for 1.0 years of catch-up. Even with rising incumbent returns ($\varepsilon_v(P^m) = 0.062 > 0$), the entrant's payoff curve is falling at the mature density ($|\varepsilon_w| = 0.037$), so mature topics are too crowded once the catch-up burden is counted (Corollary 1). Moving every topic to the peak of w raises output per researcher by 0.5% (3.3% in the within-topic calibration, whose entrant payoff curve is steeper at the mature density). The optimal frontier subsidy is 0.5% of the frontier payoff. The static welfare cost of crowding is small. If all credit lost to overlap were a transfer between racers, the gain would be 0.1% (Proposition 6). The frictionless benchmark, in which catch-up is free and topics fill to the density where $v = F$, would dissipate 12% of output per researcher. That number describes a counterfactual economy, not the calibrated one.

Writing and reading tools. Table 2 and Fig. 6 report the long-run effects of 30% improvements in each channel. Faster drafting raises output per researcher by 26.1 log points: almost one for one, as Proposition 8 predicts. It raises papers per researcher by as much and spreads researchers over more topics. Faster reading raises density by 30.7 log points and shrinks the number of topics, but raises output per researcher by only 0.23 log points. Faster catch-up is the most powerful concentrating force. Doubling catch-up speed raises mature density by 92.2 log points and lowers the overshooting threshold by 24.1 percentage points, so many more topics grow too fast. Better search lowers flagged overlap and raises the aware share. Cheaper entry leaves papers per author and density nearly unchanged and opens new topics.

Fast topics. The overshooting thresholds are 23.2% a year for myopic newcomers and 30.8% for forward-looking ones, with a sufficient bound of 38.8% (Proposition 9). These are fast but not unusual rates for hot topics on arXiv.

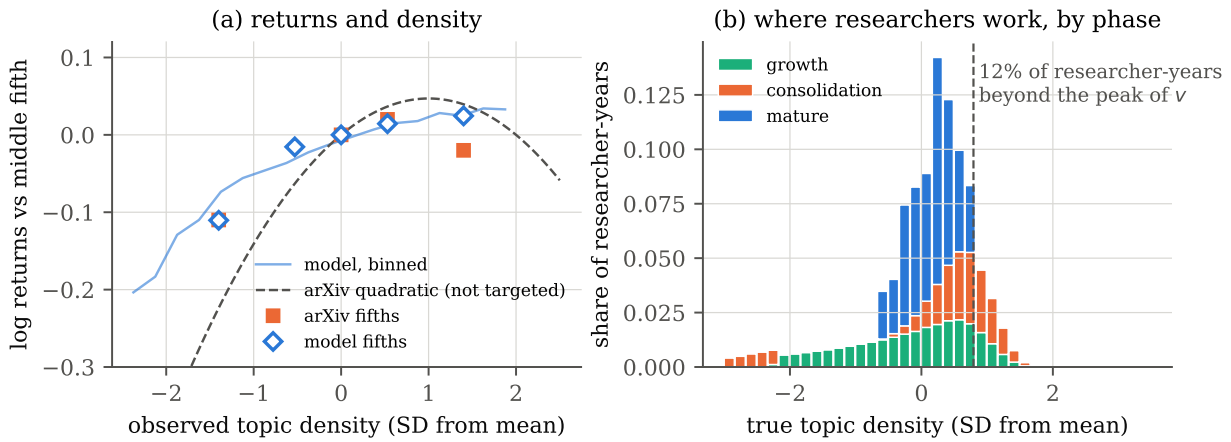


Figure 5: Returns and density in the model. Left: log returns by observed (noisily measured) topic density, relative to the middle fifth, in the calibrated cross-section, with the arXiv fifths and the arXiv quadratic (not targeted). Right: the distribution of researcher-years by true density and life-cycle phase. Researcher-years beyond the peak of v are concentrated in topics consolidating after fast growth.

Growth. Inverting (9) with $\mathcal{E} = 0.15$ gives a frontier elasticity of $\eta = 0.007$, and the stepping-on-toes parameter is $\lambda_J = 1 - \mathcal{E}|\varepsilon_w| = 0.994$. Over the arXiv range $\mathcal{E} \in [0, 0.27]$ it lies between 0.990 and one. The short-run bias from topic-age composition (Remark 4) is -0.073 over a seven-year window, so even the upper end of the long-run range keeps λ_J close to one. Topic crowding and access depletion explain little of the diminishing returns to research found by Bloom et al. (2020). Their estimates require λ_J well below one, which in our framework must come from fertility depletion (Appendix B.1) or from selection.

10 Implications: topics that grow too fast, and growth

10.1 When innovation is too fast

The stock of literature lags its flow. A topic whose literature grows fast has a small stock relative to its current density, so joining it is cheap. This subsection shows that such topics overshoot their best density, and that the result does not rely on newcomers being myopic.

Consider a topic born with flow P_0 and an empty stock. Potential newcomers learn about the topic over time. While entry pays, they join at a rate that makes the topic's flow grow at the diffusion rate g, $P(t) = P_0 e^{gt}$, and the stock follows $\dot{S} = P - \delta S$. Diffusion is the only friction on the speed of entry. Entry stops at the first date t_s at which the value of joining falls to the frontier payoff, and $P_s = P(t_s)$ is the *stopping density*. After t_s no one joins, members leave at rate ρ, and the topic thins until entry resumes at the mature density P^m. Under the fast-stock approximation of Proposition 5, entry resumes when $w(P) = F$.

Two kinds of newcomer bracket the possibilities. A *myopic* newcomer values the topic at its

Table 8: Rents, overcrowding and growth in the calibrated model

Object	Value
Peak of the payoff curve $\hat{P}$ (papers per year)	15.8
Gain from cumulative knowledge at the peak, $\hat{v}/v(P_0) - 1$	13.5%
Mature density P^m	6.2
Catch-up time $T = S/b_c$ (years)	0.99
Incumbents' premium over the frontier, $e^{\kappa P^m} - 1$	10.4%
Rent at the peak, $\hat{v}/F - 1$	14.0%
Frictionless benchmark: rent dissipated, $1 - F/\hat{v}$	12.3%
Catch-up economy: planner's gain, $\hat{w}/F - 1$	0.5%
Optimal frontier subsidy, s^o/F	0.5%
Overshooting threshold g^*, myopic entrants (per year)	0.232
Overshooting threshold, forward-looking entrants	0.308
Sufficient bound $\delta(\hat{P}/P^m - 1)$	0.388
Planner's gain if all overlap losses are transfers ($\tau = 1$)	0.1%
Short-run bias in $\mathcal{E}$ from topic-age composition (7 years)	−0.073
Slope of the entrant payoff curve $\lvert\varepsilon_w(P^m)\rvert$	0.037
Frontier elasticity η (from $\mathcal{E} = 0.15$)	0.0067
Stepping-on-toes parameter $\lambda_J = 1 - \mathcal{E}\lvert\varepsilon_w\rvert$	0.994

current payoff, so her value relative to the frontier is $v(P)e^{-\rho S/b_c}/F$. A *forward-looking* newcomer correctly anticipates that the topic will consolidate after entry stops. Her expected payoff per period over a career that starts at density P is

$$U(P) = \int_0^\infty e^{-u}\, v\big(\max\{Pe^{-u}, P^m\}\big)\, \mathrm{d}u,$$

where $u = \rho\times$ time. Her value is $U(P)e^{-\rho S/b_c}/F$.

Proposition 9 (Too fast)**.** *Let mature topics lie on the rising side of v ($\kappa\hat{P} > \ln(\hat{v}/F)$), let $w(P_0) > F$ and consider the limit $P_0 \to 0$, in which $S = P/(\delta + g)$ along the growth path.*

(i) (Myopic entry.) *The stopping density $P_c(g)$ solves $v(P)\exp(-\kappa\delta P/(\delta + g)) = F$ on its falling side. It is continuous and strictly increasing in g, with $P_c(0) = P^m$ and $\lim_{g\to\infty} P_c(g) = \bar{P}_F$, the frictionless density where $v = F$. The topic overshoots the peak, $P_c > \hat{P}$, if and only if*

$$g > g^* \equiv \frac{\rho\,\hat{P}}{b_c \ln(\hat{v}/F)} - \delta. \tag{12}$$

(ii) (Forward-looking entry.) *A topic that grows faster than*

$$g^*_{RE} \equiv \frac{\rho\,\hat{P}}{b_c \ln\big(v(P^m)/F\big)} - \delta = \delta\Big(\frac{\hat{P}}{P^m} - 1\Big)$$

overshoots the peak, $P_s > \hat{P}$. A topic that grows slower than g^* does not. Since $v(P^m) < \hat{v}$, $g^* < g^*_{RE}$.

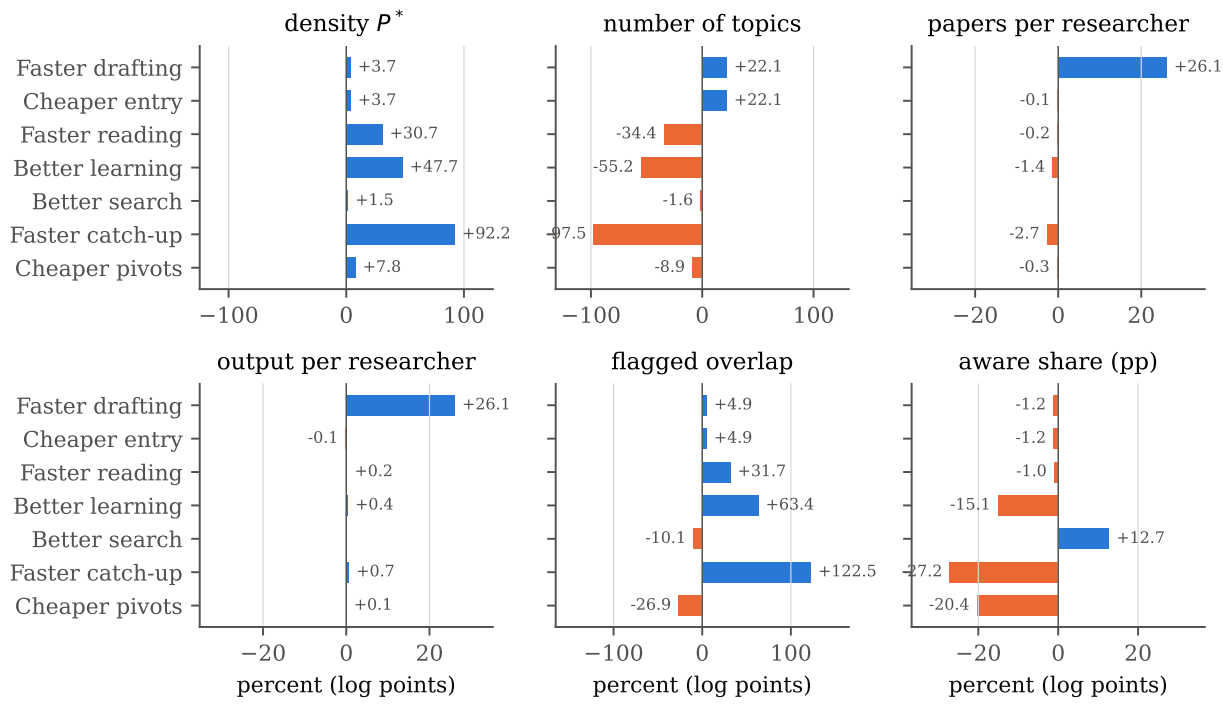


Figure 6: Long-run effects of new research tools in the calibrated model (30% shocks; catch-up speed doubled; pivot cost halved). Log points ×100; aware share in percentage points.

(iii) (Fast topics overshoot more under foresight.) *$U(P) > v(P)$ whenever $v(P)$ lies below the average of v along the consolidation path from P, which holds for every P sufficiently beyond the peak. For such topics forward-looking newcomers keep entering after myopic ones would stop: $P_s > P_c$. As $g \to \infty$, P_s converges to the root of $U(P) = F$, which exceeds the frictionless density $\bar{P}_F$.*

(iv) (Consolidation.) *After entry stops, incumbents stay, since their continuation value ($v(P_c)$ under myopia, $U(P_s)$ under foresight) exceeds F. The topic thins at rate ρ to P^m and spends $\ln(P_s/\hat{P})/\rho$ periods beyond the peak.*

The proposition gives a clean answer to the question whether "too fast" is an equilibrium phenomenon. The friction that produces overshooting is the stock lag, not myopia. Part (ii) has a simple logic. If entry stopped before the peak, the future path of the topic would lie between P^m and the peak, where v is at least $v(P^m)$. A newcomer at the stopping date would face a stock of $P/(\delta + g)$, smaller than a mature topic's P^m/δ whenever g exceeds g^*_{RE}. She would therefore earn more than a newcomer to a mature topic, who earns exactly F, and entry could not have stopped. Foresight raises the threshold a little, because a forward-looking newcomer at the peak anticipates the decline. But it deepens the overshoot of fast topics, because a newcomer to a topic far beyond its peak anticipates that consolidation will bring it back towards its best density. With instantaneous diffusion, a newborn topic with no stock fills past even the frictionless density. In the calibration the myopic threshold is 23.2% a year, the forward-looking one 30.8%, and the sufficient

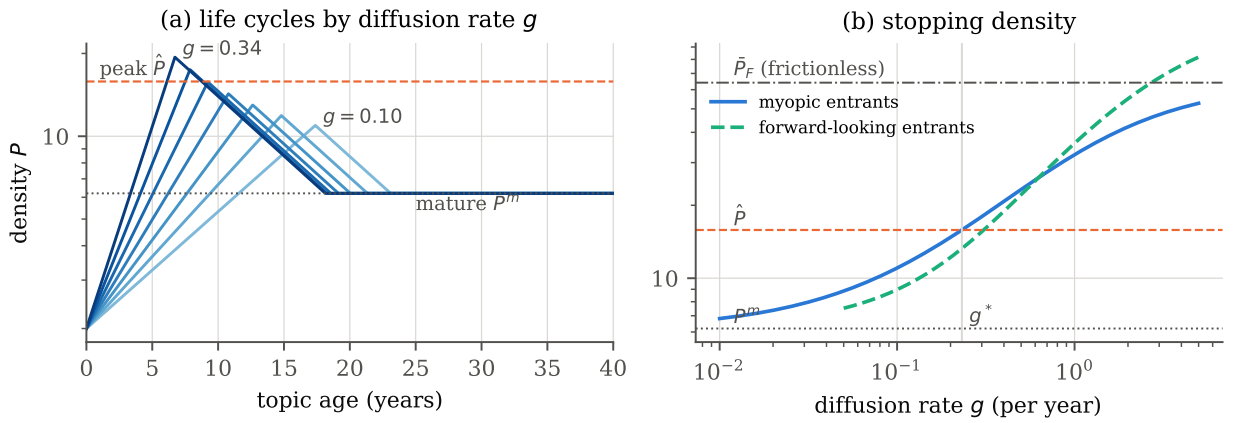


Figure 7: Topic life cycles. Left: density paths for topics with different diffusion rates g in the calibrated model; topics that grow fast overshoot the peak $\hat{P}$ and consolidate slowly. Right: the stopping density as a function of g for myopic and forward-looking newcomers, rising from the mature density P^m; with foresight, fast topics fill past the frictionless density $\bar{P}_F$.

bound 38.8%. A topic that grows at 200% a year stops at 55 papers a year with foresight, against 42 without (Fig. 7).

The result locates the overcrowded researcher-years in topics whose literature grew fast, which persist long after the rush because incumbents never pay the catch-up cost that deters newcomers. It also identifies who sets the speed limit. The thresholds fall with catch-up speed b_c, so review articles and literature-summarising tools that make catching up cheap make overshooting *more* common.

10.2 Stepping on toes, measured

Growth models since Jones (1995) let the flow of new ideas scale with researchers as N^{λ_J}, $\lambda_J < 1$, and call $1 - \lambda_J$ "stepping on toes". The parameter is reduced form. Here the flow of credited novel output is $Y = N\, a\, F(M^*(N))$, because in equilibrium every researcher's career payoff equals the frontier's. (We write λ_J to keep λ for the credit lost by an unaware duplicate.)

Proposition 10 (Stepping on toes)**.** *In the large-topic limit,*

$$\lambda_J \equiv \frac{\mathrm{d}\ln Y}{\mathrm{d}\ln N} = 1 - \eta\,\frac{\mathrm{d}\ln M^*}{\mathrm{d}\ln N} = 1 - \mathcal{E}\,|\varepsilon_w(P^*)|. \tag{13}$$

If, in addition, topics opened later are less fertile, with fertility elasticity $\eta_f \geq 0$ (Appendix B.1), the right-hand side of (13) is an upper bound on λ_J.

The formula is a sufficient statistic built from two observable objects: the volume elasticity of density, $\mathcal{E}$ (Fact 1), and the elasticity of the entrant's payoff curve at equilibrium density, $\varepsilon_w = \varepsilon_v - \kappa P^*$. Crowding does not amplify frontier depletion. It buffers it. The density margin absorbs part of the growth in researchers, so fewer new topics are opened and the frontier deteriorates

more slowly. The cost of crowding shows up in the *level* of productivity, not in its elasticity. With $\mathcal{E}$ between 0 and 0.27 and the calibrated slope of w, λ_J lies between 0.990 and one (Section 9). Topic crowding and access depletion therefore explain little of the measured decline in research productivity (Bloom et al., 2020). The rest must come from falling fertility of new topics (η_f, which density data do not identify) or from selection on researcher ability (Ekerdt and Wu, 2024).

10.3 Policy

The policy implications follow from where the rents go. Anything that raises the payoff of working inside existing topics is competed away: better reading tools, search engines, summarisers, and caps on writing. Anything that raises the frontier is not. Writing tools scale science because they scale the frontier too. The planner opens more topics than free entry does and puts each at the peak of the entrant's payoff curve. A frontier subsidy of 0.5% of the frontier payoff decentralises the optimum (Proposition 4). In practice that means credit, grants and prizes for opening new lines of work rather than for contributing to crowded ones.

11 Out-of-sample predictions

The catch-up extension was added after the returns-by-density evidence was seen, so that evidence cannot test it. The predictions below were not used to build the model. They are stated before the relevant data were analysed, and several before the data exist. To make the claim verifiable they should be deposited in a public registry before they are taken to the data. The replication package contains the text.

Predictions testable with existing data.

P1. *Overcrowding is young and fast.* The fall in returns beyond the peak should be concentrated in clusters whose flow grew faster than their stock, that is, with a high flow-to-stock ratio P/S (Proposition 9). Among mature clusters the returns curve should be monotone.

P2. *Returns follow the stock, density follows the flow.* Holding flow density fixed, returns should rise with the cluster's depreciated stock of literature (Proposition 5(ii): the premium is $\rho S/b_c$).

P3. *Newcomers start slowly where stocks are large.* The lag between a newcomer's arrival in a cluster and her first paper there, and the length of her first reference list, should rise with the cluster's stock. Incumbents should show no such gradient.

P4. *The cross-topic gradient is the catch-up premium plus audience.* The returns gradient from researchers who switch clusters should exceed the log premium of incumbents over pioneers in the clusters they join by the audience effect, which can be measured with citations from outside the cluster.

P5. *Close overlap is pinned; broad neighbourhoods fill.* The volume elasticity should rise with the similarity radius, from near zero at the overlap radius toward the fixed-field benchmark at

broad radii (Fact 1 already shows this ordering), and the ordering should survive embedding-based similarity.

P6. *Stepping on toes across areas.* Areas with a larger volume elasticity of density should show stronger diminishing returns to researchers in citation output, with $1 - \lambda_J = \mathcal{E}|\varepsilon_w|$ area by area (Proposition 10).

P7. *Search costs and detection.* Google was blocked in mainland China from 2014 (Zheng and Wang, 2020). Relative to Hong Kong-based and overseas Chinese authors in the same topics, mainland-based authors' close overlaps should have become less often aware, and their papers more often unaware duplicates, after 2014 (Lemma 1 with lower σ). The share of covering papers they pivot away from cannot be observed, so the prediction is on the aware share and on the citation penalty of their overlaps.

Predictions for the arXiv rule changes of 2025–2026.

P8. *Endorsement.* Relative to the other sections, Mathematics in 2026 and all sections in 2026–2027 relative to trend should show fewer pioneer papers (new clusters) in proportion to the fall in entrants. Local density at the overlap radius, the flagged overlap rate and the aware share should not change detectably (Proposition 7(ii) with small $\mathcal{E}$).

P9. *Review articles.* In computer-science clusters where reviews had been common, mature density should fall and the number of clusters rise relative to other computer-science clusters over 2026–2027. Fast-growing clusters there should overshoot less (higher g^*) and newcomers' first papers should come later and with longer reference lists (Proposition 7(iv), P3).

P10. *Submission cap.* The cap should leave total volume, papers per author and the overlap rate within a few percent of trend. Submissions by previously heavy submitters should be replaced partly by submissions in their co-authors' names.

P11. *Reading tools.* When literature-summarising tools are widely used, high-uptake fields where AI is not the subject should form fewer new clusters, thicken existing ones, and show more overshooting by fast-growing clusters. Overlap measured by meaning should rise in dense clusters, and citations per author-year should not rise (Proposition 7(iii)–(iv), Proposition 8).

P12. *Better search raises the aware share.* If search tools improve, the share of close overlaps that cite the earlier paper should rise, with no offsetting fall in the overlap rate in sparse clusters (Proposition 7(v)).

Two earlier predictions have already been settled. Equal returns (full dissipation) is rejected, and the catch-up cost replaces it. A detection channel that runs through the prominence of a paper's announcement is rejected by the listing-rank design (Section 8), which locates detection in deliberate search.

12 Conclusion

Is innovation too fast? The model's answer is that the question is usually asked about the wrong margin. When papers arrive faster than they can be read, researchers do not pile into the same topics until duplication explodes. They spread out, and the duplication rate stays where entry pins it. The arXiv record bears this out. Volume more than doubled while close overlap barely moved, and the entry margin responds to its cost. The costs of speed lie elsewhere: in the reading that newcomers must do to join crowded topics, in incumbents' rents behind that wall of reading, and in topics whose literature grows faster than newcomers can catch up, which overshoot even when newcomers see it coming.

The asymmetry between writing and reading is the policy-relevant result. Tools that speed up writing scale the frontier as well as existing topics, so they raise output almost one for one. Tools that speed up reading, search or catching up improve only existing topics, and entry competes the gain away unless the same tools make new topics easier to open. The calibrated magnitudes are modest. Optimal topic choice would raise output per researcher by 0.5%, and the asymmetry is large wherever the volume elasticity of density is small.

Several limits deserve emphasis. Credit is measured by citations, which also reward audience, so the rising arm of the returns profile is an upper bound on the catch-up premium. The causal test of the detection channel is a bounded null for passive visibility. It points to search costs as the right shock, and we have stated that design in advance rather than run it. The quasi-experiments on entry and catch-up costs are, so far, first stages with early reduced forms. And the frontier is a reduced-form lottery. Endogenising it, for instance by letting pioneers choose how far from existing topics to locate, is the natural next step and the margin on which the policy conclusion hinges.

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A Proofs

Throughout, "concave" means concave on the stated convex domain, and ∂_R, ∂_P denote partial derivatives.

The example satisfies Assumptions 1 and 2. The function $h(R,P) = RP/(R+P)$ is half the harmonic mean of R and P. It is concave, nondecreasing and homogeneous of degree one on $\mathbb{R}^2_+$, with $h \le \min\{R,P\}$. So $\ln(1+\beta h)$ is concave as a nondecreasing concave transformation of a concave function, and Q is bounded by $(1+\beta R)^\gamma$ for fixed R. Supermodularity: $\partial_R \ln Q = \gamma\beta h_R/(1+\beta h)$ with $h_R = P^2/(R+P)^2$ and $h_{RP} = 2RP/(R+P)^3$, so

$$\partial_P \partial_R \ln Q = \gamma\beta\,\frac{h_{RP}(1+\beta h) - \beta h_R h_P}{(1+\beta h)^2} = \gamma\beta\,\frac{2RP(R+P) + \beta R^2 P^2}{(R+P)^4(1+\beta h)^2} \ge 0.$$

For $\varphi(c) = 1 - e^{-\sigma c}$: $\varphi(0) = 0$, $\varphi' = \sigma e^{-\sigma c} > 0$, $\varphi'' < 0$.

Proof of Lemma 1. Covering papers form a Poisson count K with mean θP. Each is detected independently with probability φ, so detected and undetected counts K_d, K_u are independent Poisson with means $\theta P\varphi$ and $\theta P(1-\varphi)$. Credit is the product of $(1-\lambda)$ over undetected papers and of independent factors with mean $1-\ell$ over detected papers. Using $\mathbb{E}[t^K] = e^{-\mu(1-t)}$,

$$C = \mathbb{E}\big[(1-\lambda)^{K_u}\big]\,\mathbb{E}\big[(1-\ell)^{K_d}\big] = e^{-\theta P(1-\varphi)\lambda}\,e^{-\theta P\varphi\ell} = \exp\{-\theta P[\lambda - (\lambda-\ell)\varphi]\}.$$

Flagged overlap consists of undetected covering papers (rate $\theta P(1-\varphi)$) and detected ones that proceed (rate $\theta P\varphi q$). The aware share is the ratio of the second rate to the sum. Proceeding happens iff $\delta > 1 - \kappa_p$, which gives the conditional mean. □

Proof of Lemma 2. Let careers end at rate ρ. A newcomer to a stationary topic reads for $T = S/b_c = P/(\delta b_c)$ periods and then earns $a\,v(P)$ until her career ends. Her expected career value is $\int_T^\infty e^{-\rho t} a\,v(P)\,\mathrm{d}t = a\,v(P)e^{-\rho T}/\rho = a\,w(P)/\rho$. An explorer's value is $aF(M)/\rho$. In a stationary topic with entry flow e, the mass of members in catch-up is $e(1-e^{-\rho T})/\rho$ and the mass writing is $e\,e^{-\rho T}/\rho$. So the writing share of members is $e^{-\kappa P}$, output per member is $a\,v(P)e^{-\kappa P}$, and (1) gives $P = a\,x^*(P)\,n\,e^{-\kappa P}$. □

Proof of Theorem 1. *(i)* Since $Q(0,P) = 1$ and $\varphi(0) = 0$, $v(P) \geq g(0,P) = e^{-\theta\lambda P}$. Because Q and φ are nondecreasing in R and $R \leq b$,

$$g(r,P) \leq \bar{Q}(b)\exp\{-\theta P[\lambda - \Delta\varphi(b/P)]\} \xrightarrow[P\to\infty]{} 0,$$

since $\varphi(b/P) \to 0$ and $\lambda > 0$. At $P = 0$, $g(r,0) = 1 - r$, so $v(0) = 1$. Define $\ln C(R,0) = 0$. Then $|\theta P\varphi(R/P)| \leq \theta P$ gives joint continuity of g on $[0,1]\times\mathbb{R}_+$ ($g \to 0$ as $r \to 1$). Berge's maximum theorem gives continuity of v.

(ii) On $D = [0,1)\times(0,\infty)$,

$$\ln g(r,P) = \ln(1-r) + \ln Q(br,P) - \theta\lambda P + \theta\Delta\, P\varphi\left(\frac{br}{P}\right).$$

The first term is concave in r. The second is concave by Assumption 1 composed with a linear map. The third is linear. In the fourth, $(R,P) \mapsto P\varphi(R/P)$ is the perspective of the concave function φ, which is jointly concave on $\mathbb{R}\times\mathbb{R}_{++}$ (Boyd and Vandenberghe, 2004, §3.2.6). So $\ln g$ is jointly concave on D. Since $\ln g \to -\infty$ as $r \to 1$, the supremum over r is attained in $[0,1)$. Partial maximisation of a jointly concave function over a convex set preserves concavity (Boyd and Vandenberghe, 2004, §3.2.5). Hence $\ln v(P) = \max_r \ln g(r,P)$ is concave.

For strict concavity, take $P_1 \neq P_2$ with maximisers r_1, r_2 and $t \in (0,1)$. Under Assumption 4(a), $\partial_r \ln g(0,P) > 0$, so $r_i > 0$. If $r_1 \neq r_2$, strict concavity of $\ln(1-r)$ gives $\ln g(r_t,P_t) > t\ln g(r_1,P_1) + (1-t)\ln g(r_2,P_2)$. If $r_1 = r_2 = r > 0$, then $P \mapsto P\varphi(br/P)$ has second derivative $\varphi''(br/P)(br)^2/P^3 < 0$, which gives the same strict inequality. In either case $\ln v(P_t) \geq \ln g(r_t,P_t) > t\ln v(P_1) + (1-t)\ln v(P_2)$.

(iii) Fix ϱ with $\nu_0(\varrho) > 0$. For small P, $r = \varrho P/b$ is feasible and

$$\frac{v(P)-1}{P} \geq \frac{g(\varrho P/b,P)-1}{P} = \frac{(1-\varrho P/b)\,Q(\varrho P,P)\,e^{-\theta P[\lambda-\Delta\varphi(\varrho)]}-1}{P} \longrightarrow \nu_0(\varrho) > 0.$$

So $v > 1 = v(0)$ on some interval $(0,\epsilon)$. A strictly log-concave function is strictly quasi-concave, so its maximiser is unique. It is positive because v exceeds $v(0)$ near zero, and finite because $v \to 0$. Strict quasi-concavity gives strict monotonicity on either side of $\hat{P}$. Since v falls continuously and strictly from $\hat{v}$ to 0 on $[\hat{P},\infty)$, each $F \in (0,\hat{v})$ has exactly one root there.

(iv) For $P > 0$ the maximiser $r^*(P)$ is unique (strict concavity in r) and $\partial_P g$ is continuous. The

envelope theorem (Milgrom and Segal, 2002) gives $v'(P) = \partial_P g(r^*(P), P)$. Differentiating $\ln C$,

$$\partial_P\big[-\theta\lambda P + \theta\Delta P\varphi(R/P)\big] = -\theta\lambda + \theta\Delta\big[\varphi(c) - c\varphi'(c)\big], \qquad c = R/P,$$

which is (7). Concavity and $\varphi(0) = 0$ give $\varphi(c) - c\varphi'(c) \geq 0$.

(v) $\partial_r\partial_P \ln g = b\,[\partial_R\partial_P \ln Q + \partial_R\partial_P \ln C]$. The first term is nonnegative by Assumption 1. For the second, $\partial_R \ln C = \theta\Delta\varphi'(R/P)$, so $\partial_P\partial_R \ln C = -\theta\Delta\varphi''(c)\,c/P \geq 0$. So $\ln g$ has increasing differences in (r, P), and the unique maximiser is nondecreasing by Topkis's monotonicity theorem (Topkis, 1998). □

Proof of Proposition 1. By Theorem 1(iv), the effect of a marginal paper on an incumbent's per-period payoff is $a\,v'(P)$. Its sign is that of $\hat{P} - P$ by Theorem 1(iii). □

Proof of Proposition 2. With $\kappa = 0$, $n(P) = P/(ax^*(P))$ is continuous and strictly increasing, since x^* is nonincreasing by Theorem 1(v). Let $M(P)$ solve $\Phi(M) + Mn(P) = N$. It is unique, continuous and strictly decreasing in P (Φ is increasing). Define $\Psi(P) = \ln v(P) - \ln F(M(P))$ on $[\hat{P}, \infty)$. $\ln v$ is strictly decreasing there, and $F(M(P))$ is nondecreasing because F is nonincreasing, so Ψ is strictly decreasing. In the large-topic limit $M(\hat{P}) = N/n(\hat{P})$ and $\Psi(\hat{P}) > 0$ by assumption. As $P \to \infty$, $v \to 0$ while $F(M(P)) \geq F(M(\hat{P})) > 0$, so $\Psi < 0$ eventually. Hence there is a unique root $P^* > \hat{P}$. Crossings on $(0, \hat{P})$ violate stability, since there $v' > 0$. Payoff per researcher is $a\,v(P^*) = aF(M^*) < a\hat{v}$. □

Proof of Proposition 3. In the large-topic limit $M = N/n(P) = N\,a\,x^*(P)e^{-\kappa P}/P$, so the equilibrium condition $w(P) = F\big(Nax^*(P)e^{-\kappa P}/P\big)$ involves N and a only through Na. Log-differentiating with respect to $\ln N$,

$$\varepsilon_w\,\mathrm{d}\ln P = -\eta\,[\mathrm{d}\ln N - D\,\mathrm{d}\ln P], \qquad D = \frac{\mathrm{d}\ln n}{\mathrm{d}\ln P} = 1 - \varepsilon_x + \kappa P.$$

Solving gives (8). Then $\mathrm{d}\ln M = \mathrm{d}\ln N - D\,\mathrm{d}\ln P$, and papers per researcher $a\,x^*(P)$ move with elasticity $\varepsilon_x\,\mathcal{E}$. If $\eta = 0$, P^* solves $w(P) = \bar{F}$, which involves neither N nor a. The overlap rate and the aware share are functions of P^* by Lemma 1. Free entry into science changes P^* only through N. The general (non-limit) version replaces D by $D\,(1 - s_e)$ and $|\varepsilon_w|$ by $|\varepsilon_w|[1 + s_e(\phi_e - 1)]$, where $s_e = \Phi(M)/N$ and $\phi_e = M\Phi'(M)/\Phi(M)$. □

Proof of Proposition 4. Write $D = 1 - \varepsilon_x + \kappa P$; as above $n\,\mathrm{d}P/\mathrm{d}n = P/D$. The Lagrangian $a[f_0 M + Mn\,w(P(n))] - \mu[\Phi(M) + Mn - N]$ has first-order conditions

$$\mu = a\,w\Big(1 + \frac{\varepsilon_w}{D}\Big) \quad (n), \qquad a(f_0 + nw) = \mu\Big(\frac{1}{\pi} + n\Big) \quad (M),$$

using $\Phi'(M) = 1/\pi(M)$. Eliminating μ and using $F = f_0\pi$ gives (11). Along the resource constraint, $\mathrm{d}n/\mathrm{d}M = -(1/\pi + n)/M$, so

$$\frac{\mathrm{d}Y}{\mathrm{d}M} = a(f_0 + nw) - a\,w\Big(1 + \frac{\varepsilon_w}{D}\Big)\Big(\frac{1}{\pi} + n\Big).$$

At the equilibrium ($w = F = f_0\pi$, $\varepsilon_w < 0$) this equals $-a\,f_0(\varepsilon_w/D)(1 + n\pi) > 0$. The planner therefore opens more topics, and density falls below P^*. At $\hat{P}_w$ ($\varepsilon_w = 0$) it equals $af_0(1 - w/F) < 0$ whenever $w(\hat{P}_w) > F$. So the optimum lies in $(\hat{P}_w, P^*)$. For (ii), the left side of (11) is bounded by $(\hat{w} - F)/\hat{w}$ while $1 + n\pi \to \infty$, so $|\varepsilon_w(P^o)| \to 0$ and $P^o \to \hat{P}_w$ by strict log-concavity of w. For (iii), with a subsidy s to explorers the equilibrium condition becomes $w(P) = F(M) + s$, and $s = s^o$ makes (P^o, M^o) satisfy it. For (iv), let the planner also set x. Per-member output is $a\,g(x, P)e^{-\kappa P}$, and at fixed n, $\partial \ln P/\partial \ln x = 1/(1 + \kappa P)$. The first-order condition for x is

$$\partial_x \ln g + \big[\partial_P \ln g - \kappa\big]\frac{P}{x(1 + \kappa P)} = 0.$$

At the private optimum $\partial_x \ln g = 0$ and, by the envelope theorem, $\partial_P \ln g - \kappa = (\ln w)'(P)$. The wedge therefore has the sign of $(\ln w)'$. In the large-topic limit the planner maximises $\max_{x,P} g(x, P)e^{-\kappa P} = \max_P w(P)$. The maximum is at $(x^*(\hat{P}_w), \hat{P}_w)$, where the wedge is zero. □

Proof of Proposition 5. (i) $\ln w = \ln v - \kappa P$ is strictly concave. Since $(\ln w)'(\hat{P}) = -\kappa < 0$, the peak of w lies below $\hat{P}$. (ii) The equilibrium condition and stability are those of Definition 1. The premium follows from $v(P^m)e^{-\kappa P^m} = F$ and $\kappa P = \rho T$. (iii) $P^m < \hat{P}$ iff $w(\hat{P}) < F$, since w is falling on $(\hat{P}_w, \infty)$; this reads $\ln \hat{v} - \kappa\hat{P} < \ln F$. (iv) From $\ln \zeta + \ln v(P^m) = \ln F + \kappa P^m$, differentiate with respect to $\ln P^m$. □

Proof of Corollary 1. Total member output in a topic is $n\,a\,w(P(n))$, so one more member changes the output of the others by $n\,a\,w'(P)\,\mathrm{d}P/\mathrm{d}n$. Here $\mathrm{d}P/\mathrm{d}n > 0$ and $w' < 0$ at a stable equilibrium. Writing $w' = (v' - \kappa v)e^{-\kappa P}$ separates the spillover on incumbents from the catch-up burden. The ordering $\hat{P}_w < P^m < \hat{P}$ combines Proposition 5(iii) with stability. The loss follows from Proposition 4: per-researcher output is aF in equilibrium and tends to $a\hat{w}$ at the optimum. □

Proof of Proposition 6. (i) At private choices, $\ln s(P) = \ln g(r^*(P), P) - \theta P[(\lambda_s - \lambda) - (\Delta_s - \Delta)\varphi(br^*/P)]$. Differentiate in P. The derivative of $\ln g(r^*(P), P)$ is v'/v by the envelope theorem. The direct derivative of the bracket term is $-\theta[(\lambda_s - \lambda) - (\Delta_s - \Delta)(\varphi - c\varphi')]$. Because researchers do not maximise $\ln s$, the bracket term also changes through r^*, by $\theta b(\Delta_s - \Delta)\varphi'(c)\,r^{*\prime}(P)$. The effect of one more member on others' social output is $n\,a\,s'_w(P)\,\mathrm{d}P/\mathrm{d}n$, as in Corollary 1. (ii) With $\Delta_s = \Delta$ the bracket term is $-\theta(\lambda_s - \lambda)P$, independent of r, so $s = ve^{-\theta(\lambda_s - \lambda)P}$ and $s_w = we^{-\theta(\lambda_s - \lambda)P}$. In the large-topic limit the social planner maximises s_w, which is strictly log-concave. Its peak lies below P^m if and only if $(\ln s_w)'(P^m) < 0$, that is $(\ln w)'(P^m) < \theta(\lambda_s - \lambda) = -\tau\theta\lambda$. Since $(\ln w)'(P^m) = \varepsilon_w(P^m)/P^m < 0$, this is $\tau < |\varepsilon_w(P^m)|/(\theta\lambda P^m)$. (iii) With $\ln C_s = (1 - \tau)\ln C$,

$\ln s = \ln v - \tau \ln C^*$, and the planner maximises $\ln s - \kappa P$. □

Proof of Remark 4. Let the cross-section of topics have researcher weights $\omega(a)$ over ages a, so that the measured log density is $L = \int \omega(a) \ln P(a)\, \mathrm{d}a$. If the area's growth rate rises by $\mathrm{d}g$ for T years, the cohorts born in the window are larger by the factor $e^{\mathrm{d}g(T-a)}$ relative to older ones, so ω becomes $\omega(a)e^{\mathrm{d}g(T-a)^+} / \int \omega e^{\mathrm{d}g(T-a)^+}$. To first order, $\mathrm{d}L = \mathrm{d}g \,\mathrm{Cov}_\omega(\ln P, (T-a)^+)$. Log volume rises by $T\mathrm{d}g$ more than it would have, so the composition term in the measured elasticity is $\mathrm{Cov}_\omega(\ln P, (T-a)^+)/T$. Mature densities depend on N only through $F(M^*)$ and adjust at once, which gives (10). A constant growth rate gives a constant age composition, which area fixed effects absorb. □

Proof of Proposition 9. On the growth path, $S(t) = \int_0^t P_0 e^{gs} e^{-\delta(t-s)} \mathrm{d}s = P(t)\,\frac{1-e^{-(g+\delta)t}}{g+\delta} \leq \frac{P(t)}{g+\delta}$, with equality in the limit $P_0 \to 0$ at any fixed P.

(i) The myopic value is $w_g(P)/F$ with $w_g(P) \equiv v(P)e^{-\kappa\delta P/(\delta+g)}$, strictly log-concave. Since $w_g(P_0) \geq w(P_0) > F$, entry continues until the falling-side root $P_c(g)$. w_g is strictly increasing in g at every $P > 0$, so the root increases. The limits follow from $w_0 = w$ and $w_\infty = v$. $P_c(g) > \hat{P}$ iff $w_g(\hat{P}) > F$, that is $\ln \hat{v} - \kappa\delta\hat{P}/(\delta+g) > \ln F$, which rearranges to (12) using $\kappa\delta = \rho/b_c$.

(ii) Sufficiency, by contradiction. Suppose entry stops at $P_s \leq \hat{P}$. After t_s the topic thins to P^m, so a newcomer at t_s faces future densities in $[P^m, P_s] \subset [P^m, \hat{P}]$, where $v \geq v(P^m)$ because v is increasing below $\hat{P}$ and $P^m < \hat{P}$. Her value is at least $v(P^m)e^{-\kappa\delta P_s/(\delta+g)} \geq v(P^m)e^{-\kappa\delta\hat{P}/(\delta+g)}$. If $g > g^*_{RE}$ then $\delta\hat{P}/(\delta+g) < P^m$, so the value exceeds $v(P^m)e^{-\kappa P^m} = F$, contradicting that entry stops at t_s. Hence $P_s > \hat{P}$. Necessity of $g > g^*$: a newcomer at density $\hat{P}$ faces payoffs no larger than $\hat{v}$ at every future date, so her value is at most $\hat{v}e^{-\kappa\delta\hat{P}/(\delta+g)} < F$ when $g < g^*$. Entry therefore stops before the flow reaches $\hat{P}$. The two expressions for g^*_{RE} coincide because $\ln(v(P^m)/F) = \kappa P^m = \rho P^m/(\delta b_c)$.

(iii) $U(P)$ is the average of v along the consolidation path with weights e^{-u}, so $U(P) > v(P)$ exactly when $v(P)$ lies below that average. For $P > \hat{P}$ the path spends the stretch $u \in [\ln(P/\hat{P}), \ln(P/P^m)]$ in $[P^m, \hat{P}]$, where $v \geq v(P^m)$, so $U(P) \geq v(P^m)(\hat{P} - P^m)/P$. By the bound in the proof of Theorem 1(i), $v(P)$ decays exponentially in P, so $U(P) > v(P)$ for all P large enough. At such a P_c, the forward-looking value $U(P_c)e^{-\kappa\delta P_c/(\delta+g)} > v(P_c)e^{-\kappa\delta P_c/(\delta+g)} = F$, so entry continues: $P_s > P_c$. As $g \to \infty$ the stock vanishes and P_s is the first root of $U(P) = F$ above P^m. On $[P^m, \bar{P}_F]$, single-peakedness gives $v \geq \min\{v(P^m), v(\bar{P}_F)\} = F$ with strict inequality on the interior, so $U(P) > F$ there. Hence $P_s > \bar{P}_F$.

(iv) Incumbents have already paid the catch-up cost. At the stopping date a newcomer's value equals F after the survival factor $e^{-\rho S_s/b_c} < 1$, so an incumbent's continuation value, $v(P_c)$ under myopia or $U(P_s)$ under foresight, exceeds F, and incumbents stay. With no entry, membership decays at rate ρ. Under the fast-stock approximation $S = P/\delta$, entry resumes when $w(P) = F$, that is at P^m. □

Proof of Proposition 7. (i) and (ii) restate Proposition 3: P^* depends on Na; output per researcher is $aF(M^*)$; papers per researcher are $a\,x^*(P^*)$. Under free entry into science $F(M^*) = (\omega + c)/a$. For (iii)–(iv), $\partial_b \ln g = r[\partial_R \ln Q + \partial_R \ln C] \geq 0$. Q is increasing in β. C is increasing in σ and decreasing in ℓ, and ℓ is increasing in κ_p. w is decreasing in $\kappa = \rho/(\delta b_c)$. So w rises pointwise, which moves the falling-side root of $w(P) = F$ to the right. $M^* = N/n(P^*)$ falls because n is increasing. g^* in (12) is decreasing in b_c, and so is $g^*_{RE} = \delta(\hat{P}/P^m - 1)$ because P^m rises when κ falls. (v) A in (4) is increasing in φ, and φ is increasing in σ. □

Proofs of Propositions 8 and 10. In the large-topic limit every researcher's career payoff is $aF(M^*)$, so $Y = NaF(M^*)$ and $\mathrm{d}\ln Y/\mathrm{d}\ln N = 1 - \eta\,\mathrm{d}\ln M^*/\mathrm{d}\ln N$. By Proposition 3, $\mathrm{d}\ln M^*/\mathrm{d}\ln N = |\varepsilon_w|/(\eta D + |\varepsilon_w|)$, so $\eta\,\mathrm{d}\ln M^*/\mathrm{d}\ln N = \mathcal{E}|\varepsilon_w|$. The bound with fertility depletion is proved in Appendix B.1. For Proposition 8, $Y/N = aF(M^*)$, so $\mathrm{d}\ln(Y/N)/\mathrm{d}\omega = -\eta\,\mathrm{d}\ln M^*/\mathrm{d}\omega$, which is nonnegative because M^* falls with ω by Proposition 7(iii). For writing speed, P^* depends on Na, so $\mathrm{d}\ln(Y/N)/\mathrm{d}\ln a = 1 - \eta\,\mathrm{d}\ln M^*/\mathrm{d}\ln(Na) = 1 - \mathcal{E}|\varepsilon_w|$. □

B Extensions

B.1 Fertility depletion

Suppose topics also differ in fertility, and later topics are less fertile. The m-th topic opened has fertility $f(m) = \bar{f}\,m^{-\eta_f}$, so members of topic m earn $af(m)$ times the payoff curve. Explorers succeed with probability $\pi(M) = \bar{\pi}M^{-\eta_a}$, which captures *access depletion*: elasticity η_a, the η of the main text. A successful explorer opens a topic of fertility $f(M)$. Newcomers are indifferent between topic m and exploring when

$$f(m)\,w(P_m) = \pi(M)f(M) \iff w(P_t) = \bar{\pi}\,M^{-\eta_a}\,t^{\eta_f}, \qquad t \equiv m/M \in (0,1].$$

Proposition 11 (Scale invariance under fertility depletion)**.** *(i) If $\eta_a = 0$, the distribution of densities across topics, and hence of overlap rates and aware shares, is invariant to N and a, whatever the value of η_f. (ii) In general, with $\mathcal{D} \equiv \mathrm{d}\ln M^*/\mathrm{d}\ln N$,*

$$\lambda_J = 1 - (\eta_a + \eta_f)\,\mathcal{D}, \qquad \mathcal{D} = \Big(1 + \eta_a\,\mathbb{E}_n\Big[\frac{D(P_t)}{|\varepsilon_w(P_t)|}\Big]\Big)^{-1},$$

where $\mathbb{E}_n$ averages over topics with weights proportional to members. The paper-weighted volume elasticity of density identifies η_a, not η_f, and $\lambda_J \leq 1 - \eta_a\mathcal{D}$, the value obtained when $\eta_f = 0$. With a single density, $1 - \eta_a\mathcal{D} = 1 - \mathcal{E}|\varepsilon_w|$.

Proof. (i) With $\eta_a = 0$ the indifference condition pins P_t as a function of the relative rank t alone. Members per topic $n(P_t)$ are then invariant, so $M = N/\int_0^1 n(P_t)\mathrm{d}t$ scales with N. (ii) Every researcher's career payoff is $a\pi(M)f(M)$, so $Y = Na\bar{\pi}\bar{f}M^{-\eta_a-\eta_f}$ and $\lambda_J = 1 - (\eta_a + \eta_f)\mathcal{D}$. At

a fixed rank, $\varepsilon_w(P_t)\,\mathrm{d}\ln P_t = -\eta_a\,\mathrm{d}\ln M$. Substituting into $N = M\int_0^1 n(P_t)\mathrm{d}t$ gives $\mathrm{d}\ln N = \mathrm{d}\ln M\,(1+\eta_a\mathbb{E}_n[D/|\varepsilon_w|])$. □

Fertility depletion, the textbook "fishing out" of ideas (Kortum, 1997), thus leaves the anchor result intact and strengthens it: volume opens new topics exactly. It adds to stepping on toes a component that density data cannot see. This is why Proposition 10 delivers an upper bound on λ_J.

B.2 Fertility heterogeneity and the cross-section

With fertility ζ drawn independently at a topic's birth, a mature topic solves $\zeta w(P^m) = F$. More fertile topics are denser, their incumbents earn more, and the cross-sectional returns gradient is κP^m (Proposition 5(iv)). Topics with $\zeta\hat{w} < F$ cannot sustain replacement entry. They shrink once their growth spurt ends and are eventually abandoned. This is the model's version of research lines that die out.

B.3 Heterogeneous ability

Ekerdt and Wu (2024) attribute part of measured stepping on toes to falling researcher ability. Let ability z multiply a researcher's credit. If ability and reading are complements (Q increasing in $z\,R$), able researchers gain more from dense literatures and sort into them. If ability also raises the pioneer's success probability, they sort to the frontier. Which force wins determines whether the dense-topic premium in Fact 2 partly reflects sorting. Author fixed effects remove level differences in ability but not ability-specific returns to density. The two can be separated by interacting density with pre-period ability (early-career citations), which we leave for future work.

B.4 Credit rules

Partial credit enters only through $\Theta = \theta\lambda$ and Δ/λ. Theorem 1 requires $\lambda > \ell$ and nothing else. The sign of the externality at a stable equilibrium (Proposition 2 and Corollary 1) therefore survives whenever duplicates earn some credit. Replication value can be added as a positive social value of aware follow-ons that is not credited privately. This raises the social value of detection relative to the private one, but it does not change the sign of w' at equilibrium.

B.5 Transitional dynamics after a writing shock

A one-time increase in a raises every topic's flow immediately, before membership adjusts. Topics jump to the right along v, overlap rises, and incumbents' payoffs fall. If the jump lands beyond $\hat{P}$, the shock is crowding. Newcomers then go to the frontier and topics thin through turnover, returning to P^* at rate ρ. The overlap rate spikes and decays with half-life $\ln 2/\rho$. Papers per author rise immediately and stay higher. The author-level estimate of Fact 5, about 7% more papers for adopters, bounds the size of the writing shock so far. With adopters a minority of active

submitters (about one in fifteen in our classification), the aggregate increase in effective writing speed is at most a few percent, too small to produce a visible spike in overlap.

C Numerical methods and verification

Solving the researcher's problem. For each $P > 0$ the first-order condition $\partial_r \ln g(r, P) = 0$ has a unique root in $[0, 1)$ because $\ln g$ is strictly concave in r. We find it by vectorised bisection (60 steps) followed by three Newton steps, and set $r^* = 0$ when $\partial_r \ln g(0, P) \leq 0$. The derivative v'/v is computed from the envelope formula (7), not by differencing. Peaks and crossings use bounded Brent searches on the log-concave curves.

Stationary equilibrium and planner. The equilibrium is the solution of $\ln w(P) = \ln F(M(P))$ on $[\hat{P}_w, \infty)$. Here $M(P)$ solves $\Phi(M) + Mn(P) = N$, and the left side minus the right side is strictly decreasing (proof of Proposition 2). The constrained planner maximises output over P along the resource constraint. The full planner maximises over (P, x) by Nelder–Mead.

Forward-looking entry. The value $U(P)$ is computed by the trapezoid rule on 400 nodes of the consolidation path plus the closed-form tail at P^m. The stopping density is the first root of $\ln U(P) - \kappa\delta P/(\delta + g) = \ln F$ above P^m, bracketed on a log grid and refined by Brent's method.

Life cycles. The growth phase has closed forms, $P = P_0 e^{gt}$ and $S = P(1 - e^{-(g+\delta)t})/(g + \delta)$. The stopping time is found on a fine time grid and refined by linear interpolation. Consolidation and maturity follow Proposition 9(iii). Diffusion rates and fertility are lognormal, truncated at ± 2.5 log standard deviations and integrated on evenly spaced nodes. Because mature density is very elastic with respect to fertility ($1/|\varepsilon_w| \approx 7$), untruncated tails, where a few extremely fertile topics carry many researchers, make the moments converge very slowly. Topic-years are weighted by members, $P/x^*(P)$, and by topic survival, $e^{-\xi t}$. Observed log density adds classical measurement error with a standard deviation of 0.3, integrated by seven-point Gauss–Hermite quadrature. Besides matching the noisy arXiv proxy, this makes the quintile moments smooth in the parameters. The search uses a coarse grid (15 diffusion rates, 15 fertility levels, $\Delta t = 0.1$). Candidates are re-ranked, and all reported moments computed, on a fine grid (41 diffusion rates, 51 fertility levels, $\Delta t = 0.05$). Across grids the gradient and the share beyond the peak move by less than half a percentage point. The quintile moments move by up to one point, because the cut points between fifths fall in different places. The search bounds the median diffusion rate of new topics at 0.5 a year, so that the median topic takes at least about three and a half years to mature. With the bound at 1, both the baseline and the within-topic searches end in corners in which the median rate hits the bound, almost every topic overshoots, and the learning efficiency, catch-up speed and fertility dispersion sit at their lower bounds. The corners' losses (16.5 baseline, 12.0 within-topic) are higher than those of the interior solutions we report (14.9 and 11.4).

Verification. The file `code/tests/test_propositions.py` checks every analytical result numerically:

- log-concavity, single-peakedness, $v(0) = 1$ and $v \to 0$ on 25 random parameter draws satisfying Assumptions 1 to 4;
- the envelope formula against finite differences, and the monotonicity of $r^*(P)$ (Theorem 1);
- the initial slope $v'(0^+) = \sup_\varrho \nu_0(\varrho)$ (Assumption 4(b));
- the externality decomposition and its sign (Proposition 1);
- existence, uniqueness and the falling-side location of equilibrium for several (κ, η), and dissipation (Proposition 2);
- the volume elasticity (8) against finite differences, and exact invariance to N and a when $\eta = 0$ (Proposition 3);
- the planner's first-order condition, the ordering $\hat{P}_w < P^o < P^*$, the large-topic limit and the vanishing writing wedge (Proposition 4);
- the cross-topic returns gradient κP^m (Proposition 5) and overcrowding with rising returns (Corollary 1);
- the monotonicity and limits of the myopic stopping density, the threshold g^*, and $P_c \geq P_c^\infty(g)$ for finite birth flows (Proposition 9(i));
- under forward-looking entry, the ordering $g^* \leq g^*_{RE} \leq \delta(\hat{P}/P^m - 1)$, overshooting above the bound and none below g^*, deeper overshooting than under myopia for fast topics, and stopping beyond the frictionless density as $g \to \infty$ (Proposition 9(ii)–(iii));
- the social-value identities and the overcrowding condition under transfers (Proposition 6);
- the composition bias of the short-run volume elasticity against its covariance formula (Remark 4);
- dissipation of reading improvements when $\eta = 0$, and, when $\eta > 0$, the sign of $\mathrm{d}\ln(Y/N)/\mathrm{d}\omega = -\eta\,\mathrm{d}\ln M^*/\mathrm{d}\omega > 0$ (Proposition 8);
- the stepping-on-toes formula (13) against finite differences (Proposition 10);
- a Monte Carlo check of the credit formula (3) (Lemma 1).

All checks pass.

D Data and empirical designs

All scripts are in `empirics/` and run in numerical order. Raw data are public.

The arXiv panel. The source is the arXiv metadata snapshot distributed by Cornell University on Kaggle, file dated 18 October 2026 (Cornell University, 2026). We keep the 2,214,380 papers with new-style identifiers from January 2015 (`01_parse_snapshot.py`). For each paper we record the submitter's name, the parsed author list, the categories (the first is the primary category), title, abstract, and the timestamp of version 1, which is the submission time in UTC. The last submission in the snapshot was made at 13:59:59 ET on 15 October 2026, just before that day's cutoff.

Announcement batches and listing position (`02_listing_position.py`). Each paper is assigned the deadline it was submitted against: the same weekday's cutoff if submitted before it, otherwise the next weekday's. Weekend submissions belong to Monday's batch. The cutoff was 16:00 ET until 30 December 2016 and 14:00 ET from 2 January 2017. Both are visible as spikes in the distribution of submission times. Identifiers are assigned at announcement, so the smallest identifier among the papers with a given deadline marks the start of that announcement block. Every paper is placed in the block its identifier falls in. Papers placed in a later block than their deadline were held for moderation (6.1%). Within blocks, the rank correlation between identifier and submission time is 0.98 for papers not held (median over 3,065 blocks). Position is the rank by identifier within block and primary category. Block starts are monotone except at seven holiday periods, which we drop: 17 December to 8 January and the week of US Thanksgiving.

Cutoff design (`03_rd_sample.py`, `07_rd.py`). Weekday submissions within 180 minutes of the cutoff, deadlines from 9 January 2017 to 16 December 2022 (272,707 papers). The running variable is minutes from the cutoff. Estimates are local linear with separate slopes, a triangular kernel, category-year and weekday fixed effects, and standard errors clustered by submission day. The density of the running variable is discontinuous by construction, since authors bunch in the first seconds after the cutoff, so the main specification uses a donut of five minutes. Results for bandwidths of 30, 60, 120 and 180 minutes and donuts of 0, 5 and 10 minutes are in `empirics/results/rd.json`.

Rank design (`09_hep_sample.py`, `10_hep_pairs.py`, `11_rank_design.py`). All papers with primary category hep-th, hep-ph, hep-ex, hep-lat, gr-qc, nucl-th, nucl-ex, astro-ph.CO or astro-ph.HE, announced in their regular batch, 2017–2022, outside holidays (101,392 papers; 87,694 submitted at least 15 minutes after the batch opened). The time since the batch opened is measured from the previous weekday's 14:00 ET cutoff.

Overlap measures (`neardup.py`). Within each primary category, papers from July 2016 to December 2024 are represented by TF-IDF vectors of title and abstract: sublinear term frequencies, English stop words removed, terms in at least two and at most 30% of the category's papers. For a random sample of 3,000 papers from 2018 on, we compute the cosine similarity to the nearest earlier paper in the category within 24 months. Its percentiles define the category's thresholds τ_q. For each focal paper we keep every later paper in the category within 730 days whose similarity is

at least τ_{50}. A pair is a *self* pair if the two author lists share a name key. Half of all pairs at τ_{50} are self pairs, and at τ_{95} the share is 87%, which is why the overlap outcomes use lower thresholds than the descriptive flag.

Citations and references. INSPIRE-HEP's public API gives citation counts excluding self-citations and curated reference lists, linked by record identifiers (`inspire_fetch.py`). Coverage of the high-energy-physics sample is above 99.9%. For other fields we use the Semantic Scholar batch API (`s2_fetch.py`), whose reference lists are often withheld for arXiv papers (41% coverage among later overlapping papers). The cutoff design therefore reports all-field citations for the subsample fetched by the time of writing.

Name keys (`names.py`). Names are transliterated to ASCII, TeX accents removed, and keyed as first name plus last name for submitters and as first initial plus last name for authors (full first names where the parsed author list has them, for the rotation analysis). Name keys are used instead of database author identifiers, which were reissued for many authors in 2025. Shared names produce collisions. The designs either difference them out (endorsement, LLM adoption) or remove them explicitly (submission cap: names whose submissions in a half-month span several archives).

Endorsement (`06_entry_policies.py`). A paper has a first-time submitter if the submitter's name key has not submitted since 2015. It has a submitter with no prior authorship if, in addition, the submitter's initial key appears in no earlier author list. Weekly cells by announcement week and group (Mathematics, other sections), September 2024 to October 2026. Each week of the 2025–26 academic year is differenced against the same week of 2024–25.

CS review policy (`06_entry_policies.py`). A title is review-type if it contains survey, review, overview, tutorial, position paper, "position:", primer, roadmap, perspective, state of the art or systematic literature. The dose-response uses the 29 computer-science categories with at least 300 papers in each Nov–Oct year from 2023–24 to 2025–26. A paper has a first-time author if any author with a full first name has a key never seen before.

Submission cap (`05_cap.py`). Days 1–15 of each month, papers announced in their regular batch. A submission is excess if it is the submitter's third or later in the half-month, and clean if all of the name's submissions in that half-month fall in one archive and the submitter is not a proceedings account.

LLM-style writing (`08_llm_authors.py`). Marker words: delve, showcase, underscore (all inflections), intricate, intricacies, meticulous, pivotal, realm, notably, commendable, garner, multifaceted, seamless, invaluable, groundbreaking, tapestry, elucidate. Their share of abstracts was 2.1% in 2019, 3.0% in 2022, 10.6% in 2024 and 4.4% in 2026. The decline in 2026 suggests that authors learned to avoid them, and it is one reason to classify adoption early.

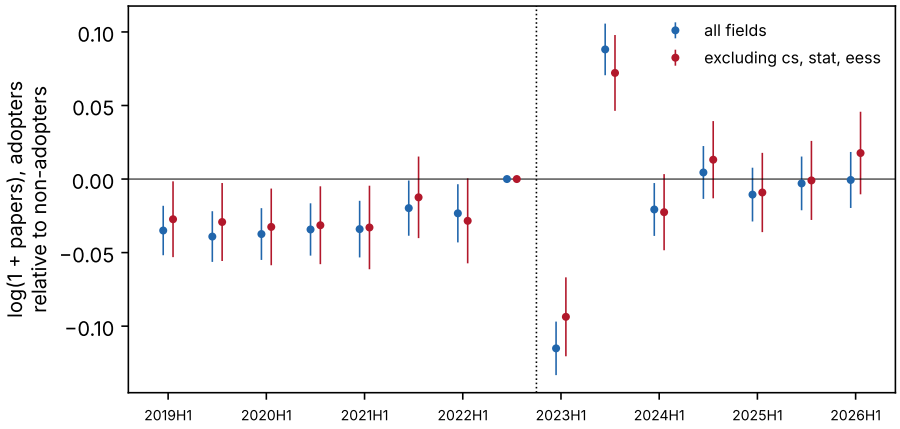


Figure 8: Author-level output of adopters of LLM-style writing, relative to non-adopters (event study, reference 2022H2; 95% intervals). Adoption is classified from 2023 abstracts. Blue: all fields; red: excluding cs, stat and eess.

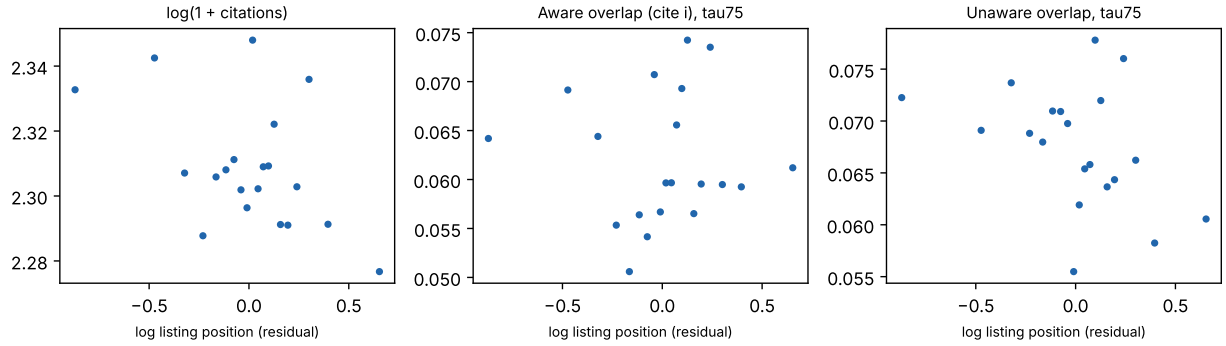


Figure 9: Listing rank in high-energy physics: binned residuals of outcomes on residual log listing position (twenty bins), net of category-year and submission-time fixed effects and log list size.